PiCovS: Pixel-Level With Covariance Pooling Feature and Superpixel-Level Feature Fusion for Hyperspectral Image Classification

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***Abstract*— In hyperspectral image (HSI) classification, convolutional neural networks (CNNs) have exhibited exceptional performance, owing to their hierarchical nonlinear modeling. However, their fixed square receptive field constrains their ability to effectively handle irregular image regions. Graph convolutional networks (GCNs) have been introduced to learn irregular regions through correlations between adjacent pixels modeled as superpixel-based nodes, yet they lack pixel-level information. We propose a novel approach Pixel-level with Covariance Pooling feature and Superpixel-level feature Fusion for HSI Classification (PiCovS). Our method harnesses complementary spectral–spatial features at both pixel-level and superpixel-level to capture characteristics of both small-scale regular and large-scale irregular regions. We introduce a hybrid network that integrates and propagates features between image-level pixels and graph-level nodes using a graph encoder–decoder, effectively reconciling the differences between regular CNN and irregular GCN data representations. To enhance superpixel boundary learning, we modify the manifold simple linear iterative clustering (M-SLIC) algorithm by incorporating texture feature information, resulting in refined superpixel representations. In addition, we propose a novel covariance pooling mechanism with an attention mechanism within the CNN branch, enabling the capturing and utilization of holistic HSI information along spectral and spatial dimensions by exploiting second-order statistics throughout the network. Our comprehensive experiments showcase the efficiency and robustness of the proposed framework, achieving an impressive overall accuracy of 99.84%, 99.97%, 99.98%, and 81.96% on the Indian Pines, University of Pavia, Salinas, and Houston University datasets, respectively. Remarkably, PiCovS excels even with limited training samples, outperforming other state-of-the-art methods in accuracy.**

***Index Terms*— Convolutional neural network (CNN), covariance pooling, feature fusion, graph convolutional network (GCN), hyperspectral image (HSI) classification.**

I. Introduction

HYPERSPECTRAL image (HSI) classification is the automation of categorizing land-cover types through the use of machine learning algorithms. It has since been applied in agricultural and land resources assessment, geological exploration, and pollution monitoring for purposes, such as environmental protection, military reconnaissance operations, and disease diagnosis. Earlier works have mainly focused on investigating ways of establishing mapping functions and hyperplanes search for dimensionality reduction of spectral bands using conventional machine learning methods, such as logistic regression [1], extreme learning machine [2], and kernel-based method [3]. However, these conventional machine learning techniques are dependent on features engineered by human experts leading to a high margin of errors and nonapplicability in real-world cases.

Deep learning methods [4], [5] have remarkably propelled the studies and development of HSI classification methods [6], [7], [8]. Their ability to learn nonlinear and high-level features robustly in a hierarchical manner has revolutionized the way HSIs are classified. Convolutional neural networks (CNNs), deep belief networks (DBNs), and recurrent neural networks (RNNs) have been the most widely applied deep learning methods for HSI classification [9], [10]. Among them, CNNs have been widely used and implemented in different forms, including single-based architectures [11], [12] and hybrid-based architecture [13]. However, CNNs are mostly designed for Euclidean data, which ignores the correlation between land covers given their square filter receptive sizes and their inherent nature of modeling regular spatial structures, making them unable to properly capture and model irregular regions and the internal connections of various land covers present in HSIs [8]. They also require higher computing capacity and longer training with huge optimization parameters to generally learn land covers, especially along the spectral bands and spatial dimensions [10].

Graph convolutional networks (GCNs) [8] have received increased attention, given their ability to perform convolutions on structured graphs. GCNs have been used to exploit the correlation between adjacent land covers and encode HSIs into structured graphs to provide spatial contextual structures [8], [14], [15]. GCNs use superpixel-level features to aggregate pixels to capture the contextual multiscale spatial information by constructing an adjacency matrix. However, the number of pixels in HSIs is enormous; therefore, utilizing each pixel as a graph node would require high computational resources, which will limit the applicability of modeling HSI with GCNs

Manuscript received 1 August 2023; revised 5 September 2023; accepted 8 September 2023. Date of publication 6 October 2023; date of current version 23 October 2023. This work was supported in part by the National Natural Science Foundation of China, Major Instrument Project the Development of Fetal Heart Sound-Cardiac-Ultrasound Multimodal Auxiliary Diagnostic Equipment, under Grant 62027827. *(Corresponding author: Obed Tettey Nartey.)*

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Digital Object Identifier 10.1109/TGRS.2023.3322641

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[16]. To alleviate this challenge and exploit the power of GCNs for HSI classification, Wan et al. [17] proposed to use superpixel centroids as nodes to reduce the size of the graph structure and the computational complexity to train models of non-Euclidean data representations in an end-to-end manner. Qin et al. [14] proposed a mechanism to construct the adjacency matrix by dividing subgraphs and dynamically changing the weight between nodes. The method enables specifying different weights to different nodes in a neighborhood, not depending on having prior knowledge of the graph structure. A spectral–spatial GCN [17] was devised to extract features by simultaneously considering the spectral and spatial neighborhoods of each pixel. Although this mechanism helps to alleviate the issue of small-sample sizes, GCN-based models fail to capture small regular spectral–spatial contextual information at the pixel-level by the graph nodes and superpixel modeling of irrelevant pixel-level information leading to incorrect calculation of fixed edge weights and limit the expressive nature of the network. It leads to misclassification in the classification map. Furthermore, exiting superpixel GCN methods only captured and propagated information between adjacent superpixels while ignoring the local spectral–spatial information within superpixels, which is essential for the classification of HSI with many small-scale objects. Duan et al. [18], propose a structure-preserved hyper-GCN (SPHGCN) to mitigate GCN methods that consider the simple point-to-point structure between two pixels rather than the high-order structure of multiple pixels, contradicting real feature distribution of land covers. Zhou et al. [19] improved the work by Duan et al. [18] by proposing attention multihop graph and multiscale convolutional fusion network (AMGCFN), including twin subnetworks of multiscale fully CNN and multihop GCN to extract the multilevel information of HSI to make good use of the advantages of CNN and GCN. However, these models may improve models' computational efficiency, in terms of accuracy, time, and computational resources, Very few have done their studies on a small sample set and obtained similar outputs. Zhang et al. [20] have proposed to utilize graph aggregation for cross-domain few-shot learning to achieve generalization. They tackled the issue of small labeled data problems using few-shot learning. Their work was improved by Zhang et al. [21] by including text information containing extensive prior knowledge about land cover classes to explore the effectiveness of linguistic mode in HSI classification. Cross-domain adaptation [22], [23], [24] techniques have been employed owing to the small nature of datasets available and obtained remarkable performance. Zhang et al. [25] proposed a structural optimization transmission framework for collaborative land-cover classification of HSI and light detection and ranging (LiDAR) data. Their method fully leverages the reflectance-specific information of HSI and the detailed edge (structure) representations of multisource data.

Owing to the aforementioned problems, hybrid methods that fuse information have been proposed [26], [27], [28]. Zhang et al. [26] proposed information fusion for the classification of hyperspectral and LiDAR data that fused heterogeneous information to improve the classification performance of HSI and LiDAR data. Some studies also combined the characteristics of CNNs and GCNs to design methods for HSI classification. Liu et al. [29] proposed a CNN-enhanced GCN (CEGCN) with pixel- and superpixel-level feature fusion (CEGCN) to capture and fuse superpixel-level and pixel-level for HSI classification obtaining high performance. Dong et al. [10] improved the performance accuracy by proposing the weighted feature fusion of CNN and graph attention network (WFCG), which combines the characteristics of CNN and graph attention networks (GATs). In their work, they used graph encoder and decoder modules to connect the grid and nodes, which can better combine the features of CNN and GAT to better capture long-range information and high-level features. However, these models only allow for the long range information propagation between adjacent superpixels and in the process, ignore the spectral–spatial information. To alleviate the problem, Zhang et al. proposed an end-to-end mixhop superpixel-based GCN (EMS-GCN) [30]. The authors introduced a differential superpixel segmentation algorithm followed by a mixhop GCN, which is designed to extract both the local information within superpixels and long-range information among superpixels while, at the same time, updating the structure of the superpixel graph. Other learning and remote sensing methods such as transformer-based networks [7], [25], [31], [32], [33] and attention-based networks [34], [35], [36] have given new insights into HSI classification. All of these studies supply a broad prospect for the application of CNNs in large-scale HSI classification of land covers [27], [28], which has significantly promoted the research of this field, yet these methods primarily employ simple first-order statistics of deep features to represent HSIs, ignoring characterize complex global feature distribution (high-order statistics) and also did not explore their methods on a small sample size given the scarcity of labeled data leading to limitation of classification performance and larger computational cost.

Recently, high-order statistical representations [37] have shown advantages in obtaining deep feature statistics of visual images, which have attracted more attention in large visual image recognition tasks [38], [39], [40]. In direct comparison with first-order statistical CNN representations, high-order statistical models can capture complex feature relationships via higher order functions, integrated as part of CNN layers. The pioneering works, such as bilinear CNN (B-CNN) [39], [41] and covariance pooling [42], [43], [44], performed global second-order pooling operations focusing on small-scale scenarios. He et al. [45] had proposed multiscale covariance maps and applied them to 2-D CNN to fully exploit the spectral–spatial information presented in HSI. Zhang et al. [43] then designed a two-branch CNN network to construct and cluster neighboring local pixels via a covariance matrix between different spectral bands. Given the arduous optimizations needed for second-order pooling mechanisms, Xue et al. [42] proposed an attention-based second-order pooling to optimize and capture the inherent spatial information of HSIs. However, all the aforementioned methods only included the high-order statistical function after the last convolutional layers in an end-to-end manner, which

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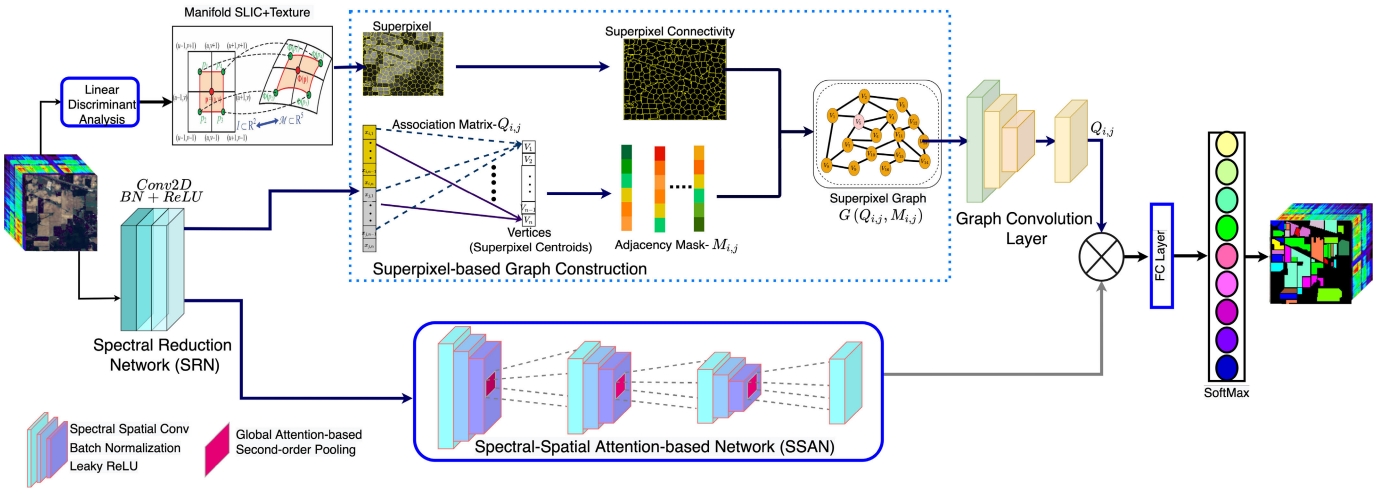


Fig. 1. Schematic illustration of the proposed pixel–superpixel-level feature fusion-based second-order pooling network (PiCovS) for HSI classification.

limits the use of holistic image information across a network. To better characterize complex boundaries of HSIs with several classes in a very high-dimensional latent space, one possible solution is to learn higher order representations to improve the nonlinear modeling capability of CNNs from lower to higher layers.

In view of the problems aforementioned and motivated by nonlocal networks used in part with squeeze-excitation networks (GCNet) [36], this article proposes a pixel–superpixel-level feature fusion with second-order pooling network [Pixel-level with Covariance Pooling feature and Superpixel-level feature Fusion for HSI Classification (PiCovS)], which fuses the CNN and GCN into an end-to-end hybrid deep-learning network for HSI Classification. Considering the strengths of CNN to learn small-local spectral–spatial features at the pixel-level and GCN to model the various spatial structures of land covers as graph nodes, we propose to effectively exploit the spectral–spatial information by introducing a modified manifold superpixel generation algorithm that incorporates image texture along with the color and spatial features when measuring the similarity of pixels to generate superpixel boundaries and better preserve information within superpixels. This way enables the propagation of features between image-level pixels and graph nodes for HSI classification.

Second, to address the limitation of existing models in capturing high-order statistical features, we introduce a new second-order representation mechanism with an attention module. This mechanism spans from lower level layers to higher level layers in the network, allowing the network to effectively capture second-order statistical holistic features. As shown in Fig. 1, the GCN branch (see Fig. 1 (top), yellow rectangle), a graph encoder–decoder is designed to propagate features from image pixels to graph nodes and vice versa. The CNN part of the proposed framework learns features at the pixel-level on the small regular regions, while the GCN branch models the large irregular regions of HSI and generates its superpixel-level features.

Initially, a transformation technique is implemented on HSIs to reduce irrelevant spectral information to decrease the computational cost of following operations and enhance significantly contributing spectral features that are robust and discriminative. Then, the large-scale spatial structure of HSIs is modeled into graphs to generate superpixel-level features using a modified manifold simple linear iterative clustering (M-SLIC) [46] method. M-SLIC, despite its fast computational time compared with other superpixel generation algorithms, ignores the spectral similarity of neighboring pixels when merging small clusters to form large clusters. The texture of pixels is also not taken into consideration by the M-SLIC technique when computing the similarities of pixels, a feature that is so essential in high-spatial resolution remote sensing images. Therefore, we incorporate texture feature information into the M-SLIC to enhance the superpixel generation process and later feed into the superpixel-based GCN, as shown in Fig. 1.

To enhance the feature learning capability of our model, we propose the incorporation of a second-order statistics module with an attention mechanism within the CNN part of our proposed method. This module is designed to capture holistic HSI information by producing a covariance matrix to compute pairwise correlations of features. By doing so, we aim to model more discriminative and representative features along both the spectral and spatial dimensions. The utilization of the inner product computation to derive pairwise correlations allows the nonlocal operation to effectively capture feature dependencies even across distant positions. Consequently, this nonlocal operation accentuates the most influential features while downplaying the less significant ones, aligning with the principles of the self-attention mechanism. Through this innovative self-attention second-order statistics mechanism, diversity is introduced among neighboring pixels, facilitating the capture of long-range dependencies along the spectral–spatial dimension. As a result, the self-attention mechanism is adapted to directing attention toward the most pertinent channels, effectively emphasizing significant features.

The incorporation of this self-attention second-order statistics mechanism enhances the capability of our proposed method to capture long-range dependencies, emphasize

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significant features, and produce more discriminative representations along both spectral and spatial dimensions, ultimately contributing to the success of our proposed approach in HSI classification. This, in turn, leads to improved feature representations and enhanced performance in the classification of HSIs.

The main contributions of our work can be summarized as follows.

1) We propose an end-to-end hybrid network for HSI classification, which merges pixel-level high-order statistical features and superpixel-level features. It utilizes a graph encoder and decoder to connect graph grids and nodes into image pixels to effectively propagate features within the network enabling the model to capture both regular and irregular spectral–spatial patterns.
2) In the superpixel-based GCN, we modify the M-SLIC to incorporate texture feature information to better learn superpixel boundaries to generate refined superpixel representations for the superppixel-based GCN.
3) We design a high-order statistical context block and introduce it within network layers to ensure network training stability. It utilizes a correlation matrix to capture holistic HSI information, enhancing model learning for greater representativeness and discriminative power.
4) We propose a novel attention mechanism in the pixel-level branch. This mechanism captures extensive long-range information and high-level features from the high-order statistical context block. By prioritizing pertinent data, the model's discriminative capability is enhanced, which contributes to improved classification performance.

II. Pixel-Level With Covariance Pooling Feature and Superpixel-Level Feature Fusion

HSI classification involves the classification of pixels with labeled classes. Letting $\mathbf{X} \in \mathbb{R}^{N\times H\times W\times B}$ be HSIs with B bands and spatial dimension of height H and width W; the total number of pixels to be classified is denoted as N having a total number of C classes. We denote the dataset as $D = \{X, Y\}_{n=1}^{N}$, where Y is the label for the image X containing N pixels. The primary objective of this research is to perform accurate pixel-level classification in HSIs. To achieve this goal, we propose an integrated approach that combines three key components: GCN, pixel-level CNN, and a second-order statistics attention scheme. By leveraging these components, our method can effectively classify HSIs in a supervised learning setting. As depicted in Fig. 1, the work comprises three key components, each contributing to the overall effectiveness of the approach. *Superpixel-Based Graph Network (SGN):* This component constructs and models superpixels, and generates superpixel-level features. By leveraging the spatial information within superpixels, the SGN efficiently captures local contextual relationships, enhancing the representation of spatial features. The spectral reduction network (SRN) reduces nonsignificant information while enhancing the salient spectral features in the hyperspectral data. This reduction process helps to focus on relevant spectral information, facilitating better discrimination between classes. The spectral–spatial attention-based network (SSAN) extracts pixel-level feature maps, effectively capturing both spectral and spatial features of individual pixels. This approach is enriched by the inclusion of a second-order module, which extracts high-order features through a self-attention mechanism. This novel addition allows the model to capture complex dependencies in the hyperspectral data, leading to improved performance in HSI classification tasks, further enhancing the discrimination capability of the model. Finally, to complete the classification process, a softmax classifier is employed, guiding the training process and enabling the model to classify the extracted features into distinct classes.

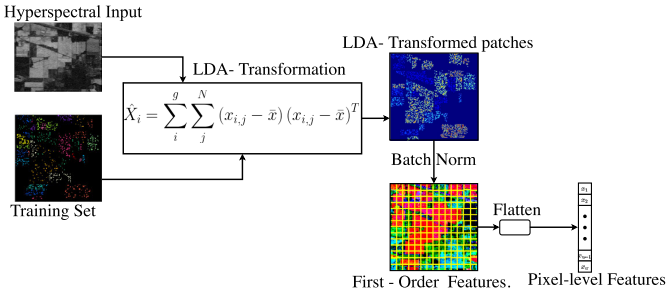


Fig. 2. Graphical illustration of the dimensionality reduction process by using the LDA method.

A. Superpixel-Based Graph Network

1) Superpixel Generation: Superpixels are generated to allow GCN to model the clustered pixel centroids as nodes. Unlike CNNs, which can model pixels from images, GCNs model superpixel nodes to construct suitable graph structures that consist of information that has been extracted along the spectral and spatial dimensions. As it is provided in Fig. 2, the linear discriminant analysis (LDA) is employed as the dimensionality reduction technique to preprocess and reduce the dimensions of $x_i \in X$, after which the M-SLIC [46] method is applied to generate superpixels of HSIs, as shown in Fig. 3. M-SLIC splits the image into $S = ([(H \times W/\rho)])^{1/2}$ several spatially connected and similar superpixels to reveal the spatial structure. The superpixel segmentation scale $\rho = \rho \geq 1$ is used to control the average size of superpixels to be generated. However, M-SLIC, despite its fast computation time, has two major drawbacks. M-SLIC does not consider the spectral similarity when merging small clusters into their largest neighbor cluster, and it also ignores texture, which is one of the most important features of high spatial resolution (HSR) remotely sensed images when measuring the similarity of pixels

$$D_{CS,ji} = \sqrt{\left(D_{C,ji}\right)^2 + \left(\frac{D_{S,ji}}{s}\right)^2 m^2} \tag{1}$$

where $D_{C,ji} = ((l_j - l_i)^2 + (a_j - a_i)^2 + (b_j - b_i)^2)^{1/2}$ and $D_{S,ji} = ((x_j - x_i)^2 + (y_j - y_i)^2)^{1/2}$ are the color similarity distance and the spatial proximity distance between the j_{th} cluster center and the i_{th} pixel, respectively. Equation (1) computes the feature distance to assign each pixel that is

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labeled by the most similar cluster within a search window of $2S \times 2S$ pixels. m is the compactness coefficient, which is used to weigh the relative significance between color similarity and spatial proximity.

To incorporate texture into M-SLIC to solve the shortcoming M-SLIC of ignoring texture, which is essential in HSR, we leverage the gray-level co-occurrence matrix (GLCM) [47] to extract texture features based on second-order statistics in calculating the correlation between two gray levels to reflect spatial-structure information about directions, intervals, and changes in magnitude and speed. GLCM is a matrix where the number of rows and columns is equal to the number of gray levels, G, in the HSI. We adequately extract the texture features by utilizing $(\Delta x, \Delta y)$, with $(0, 1)$, $(1, 1)$, $(1, 0)$, and $(-1, 1)$ as the four types of separating distances. Since M-SLIC impacts the feature distance where the greater the distance, the less similar the pixel is to the cluster center, we inverse the texture matrix according to the following equation:

$$\mathrm{Mat}_T = (\max(\mathrm{Mat}) - \mathrm{Mat})/\max(\mathrm{Mat}) \tag{2}$$

where Mat_T and Mat are the inverse and original texture matrices, respectively.

Therefore, to incorporate the texture information in the M-SLIC, the 5-D vector $[l, a, b, x, y]^T$ in the M-SLIC is turned into $[C, t, x, y]^T$, where $[l, a, b]^T$ is the color feature vector in the CIELAB color space. $[C]$ is the color feature vector, and $[t]$ is the texture feature vector. $[C]$ is then normalized, and then, the distance between each pixel i_{th} and the cluster center j_{th} is given using the formulation in the following equation:

$$D_{CTS,ji} = \sqrt{\left(D_{CT,ji}\right)^2 + \left(\frac{D_{S,ji}}{s}\right)^2 m^2}. \tag{3}$$

From (3), $(D_{CT,ji}) = (D_{C,ji}^2 + D_{T,ji}^2)^{1/2}$ is the combined distance of the color and texture features $D_{C,ji}$ $D_{T,ji}$, respectively. In detail, the superpixel generation process involves the following steps.

1) *LDA Preprocessing:* We apply LDA to the HSI to reduce its dimensionality while retaining the most discriminative spectral information. This step helps in transforming the original HSI data into a lower dimensional space.
2) *M-SLIC Superpixel Generation:* After the LDA preprocessing, we use the M-SLIC algorithm to partition the LDA-transformed HSI into a set of superpixels. The M-SLIC algorithm groups pixels that have similar spectral properties and are spatially adjacent, creating compact and connected regions.
3) *Spatial Structure Revealing:* The M-SLIC-generated superpixels reveal the spatial structure of the HSI, as pixels within each superpixel share similar spectral properties and are spatially connected. This spatial structure information is essential for capturing contextual relationships and improving the performance of subsequent processing steps, such as feature extraction and classification.

By combining LDA preprocessing and M-SLIC-based superpixel generation, our method effectively reduces the dimensionality of the HSI and captures the spatial contextual information, leading to improved feature representations for HSI classification. The superpixel generation process ensures that pixels with similar spectral and spatial characteristics are grouped together, which enhances the efficiency and accuracy of subsequent analysis tasks. Thus, the node is expressed as $V = [v_1, v_2, v_3, \ldots, v_j]$, where V is the total feature vector.

2) Graph Generation: An undirected graph is defined as $G = (V, E)$, where V: matrix of the feature vectors and E denote the nodes and edges of the graph, respectively. In this case, the superpixels of an HSI are used to construct G. The nodes $V = [v_1, v_2, v_3, \ldots, v_j]^T$ contain the computed centroids. The adjacency relations between superpixels in an HSI are used to construct E. Letting Q be the mapping association matrix of pixels and their corresponding superpixels, Q is obtained by using (4), where $\forall \tilde{n} \in \mathrm{Flatten}(x_i)$ making $Q_{i,j} \in \mathbb{R}^{hw \times S}$ be the relational mapping value of the nth pixel in the jth superpixel

$$Q_{i,j} = \begin{cases} 1, & \text{if } \tilde{n}_j \in S_j \\ 0, & \text{otherwise.} \end{cases} \tag{4}$$

The undirected graph G is then constructed by encoding x_i into nodes according to $Q_{i,j}$, thereby making the nodes V of the undirected graph to be reformulated as

$$V = \mathrm{Encode}\left(x_i, Q_{i,j}\right) \tag{5}$$

which can conveniently be convoluted by the GCN. A graph decoder is then constructed to propagate graph node features to pixel features along with Q using (6) for classification

$$\hat{x}_i = \mathrm{Decode}\left(v_j, Q_{i,j}\right). \tag{6}$$

This is constructed because the pixel features are used for classification instead of the generated superpixels. The features obtained after the decoding process are then projected to image space by reshaping the features to restore x_i to its original input spatial dimension before the flattening operation is performed on it. The adjacency was constructed by following the work done by [29]. Specifically, the connectivity between nodes, which is determined by sharing a common boundary, is encoded to form the adjacency mask by the following equation:

$$M_{i,j} = \begin{cases} 1, & \text{if } \tilde{S}_j \text{ and } S_{j+1}, \text{ are adjacent} \\ 0, & \text{otherwise.} \end{cases} \tag{7}$$

From (7), $M_{i,J}$ is the adjacency mask of the i_th HSI with j number of superpixels. This records the adjacency that exists between covers in HSIs to determine the edge E of the unweighted graph, as demonstrated in Fig. 3. In brief, we construct an unweighted graph from an HSI by utilizing the superpixel segmentations generated. The encoder–decoder network is able to project image-level features to the graph-level feature spaces and vice versa, which is conveniently plugged into the network.

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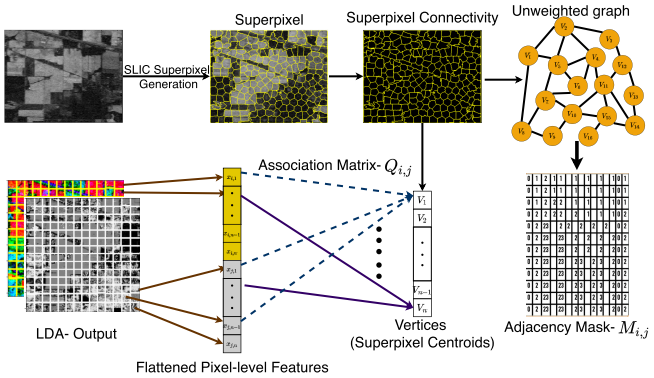


Fig. 3. Schematic of the superpixel generating module. Taking an HSI as input, an LDA-SLIC superpixel generating method is used to generate superpixels. The superpixels are used as nodes to construct an unweighted graph, where the vertices of the nodes are the superpixel centroids. x_i and x_j denote the pixel-level features mapped to the superpixels centroids v_i.

3) Graph Convolution Network: Given a graph $G = (V, E)$, the vertices V are encoded into a node matrix Π and adjacency matrix $\mathcal{M}$, where the i_th row is node i and the adjacency matrix $\mathcal{M}_{i,j}$ denote the edge weight of the connection between i_th node and node j. Following the work by Shuman et al. [48], we performed spectral decomposition based on the Laplacian matrix graph computation. We, therefore, define the graph convolution operation as

$$y = U_{d\theta}(\lambda)U^T X. \tag{8}$$

From (8), y, U, and X are the output, the orthogonal matrix, and input vectors, respectively. U consists of the feature vectors in the Laplacian matrix $L = D^{-(1/2)}(D - \mathcal{M})D^{-(1/2)}$. $d\theta(\lambda) = \text{diag}(\theta_1, \theta_2, \ldots, \theta_n)$ is the learned parameters. It should be noted that D denotes the degree for matrix $\mathcal{M}$. With a large number of parameters to be learned, (8) is further improved by computing $\theta_i = \sum_{j=0}^{S} \alpha_j \lambda_i^j$. λ_i^j is the j_th power of L's i_th eigenvalue, and α_j denotes the convolution parameter to be learned. Setting the size of the receptive field S to 1, the graph convolution operation in (8) changes to

$$y = \alpha_0 X + \alpha_1 L X \tag{9}$$

$$= \beta\left(I + D^{-\frac{1}{2}}\mathcal{M}D^{-\frac{1}{2}}\right)X \tag{10}$$

where $\beta = 0.5\alpha_0$, $-\alpha_1 = 0.5\alpha_0$, and $I + D^{-(1/2)}\mathcal{M}D^{-(1/2)}$ denotes the j_th power of L. The identity matrix I is added to ensure the stability of numerical values when iterating the convolution operation a number of times. The edge weight is then computed from the input features of each layer to improve the adaptability of G. Letting the output of the l_th layer be Π^l, the adjacency matrix $\mathcal{M}^l$ in l would be computed as follows:

$$\mathcal{M}^l = \sigma\left(\left(\text{BN}(\Pi)^{l-1}W_\phi^l\right)\left(\text{BN}(\Pi)^{l-1}W_\phi^l\right)^T\right) \cdot M + I. \tag{11}$$

$\text{BN}(\Pi)^{l-1}$ denotes the batch-normalized input feature of the l_th layer, and W_ϕ^l denotes the parameters to be learned. Then, the Hadamard product $(\cdot)$ is computed from the adjacency mask M and an identity matrix, as provided in (11). The nonlinear function σ compresses the edge weights between 0 and 1. In this work, σ is a sigmoid function that goes to ensure that there is numerical stability. The adjacency mask M is used to ensure that the connectivity between adjacent nodes is kept while cutting off any association of nonadjacent nodes. Considering this, the graph convolution operation is finally formulated as

$$\Pi^l = f\left(\left(D^l\right)^{-\frac{1}{2}}\mathcal{M}^l\left(D^l\right)^{-\frac{1}{2}}\text{BN}(\Pi)^{l-1}W_\phi^l\right). \tag{12}$$

Similarly, from (12), D^l is the degree matrix of the adjacency matrix A^l. Equation 12 models spatial structures of land covers on graphs to generate superpixel-level features Π^l. Letting GCN consist of l_s where $1 < l_s$ layers, the input, and output features are Π^0 and Π^{l_s}, respectively. The module is then given as $\Pi^{l_s} = \text{GCN}(\Pi^0, M)$.

B. Spectral Reduction Network

In HSIs, each pixel contains a large number of spectral bands, which often include noisy and irrelevant information. This redundant spectral information can negatively impact the performance of HSI classification models, leading to decreased accuracy and efficiency. The SRN module is specifically designed to address this issue. The SRN performs a dimensionality reduction operation on the spectral information by selectively retaining only the most discriminative and informative spectral features. By filtering out irrelevant and noisy spectral bands, the SRN aims to improve the overall classification accuracy and build a more robust HSI classification model. Given a tensor of a feature map $X^l \in \mathbb{R}^{h' \times w' \times b'}$ as an input to the l_th layer in the spectral convolution block, the spatial location $S_0 = (x, y)$ for each output feature map is computed as

$$X_i^l(S_0) = f\left(\mathcal{W}_i^l \cdot \text{BN}(X)^{l-1}(S_0) + \beta_i^l\right) \tag{13}$$

where $\text{BN}(X)^{l-1}$ is the batch-normalization operation on the input feature map of the l_th layer with $\mathcal{W}_i^l$ and β_i^l kernels and biases, respectively. The inner product $(\cdot)$ between the batch-normalized and the kernel vectors is computed, which, when added to the bias, is activated by an activation function $f(*)$ to produce $X_i^l(S_0)$. $X_i^l(S_0)$ is the output feature map of the j_th channel at the spatial location (S_0). Letting SRN be made up of L_{sc}, the module can be simply formulated as $X^{L_{\text{sc}}} = \text{SRN}(X^i)$, where X^l and $X^{L_{\text{sc}}}$ denote the input and the output, respectively.

While the SRN primarily focuses on reducing the dimensionality of the spectral domain, it indirectly benefits the spatial domain as well. By eliminating irrelevant spectral information and noise, the SRN contributes to cleaner and more informative spectral–spatial features, which are crucial for accurate HSI classification

C. Spectral–Spatial Attention-Based Network

1) Covariance Pooling Normalization: The SSAN with covariance pooling is proposed to make use of diverse neighboring pixels by extracting deep spectral and spatial features to enhance useful information and suppress the effects of less-contributing information. In the proposed context block, we adopt a covariance pooling to compute the second-order

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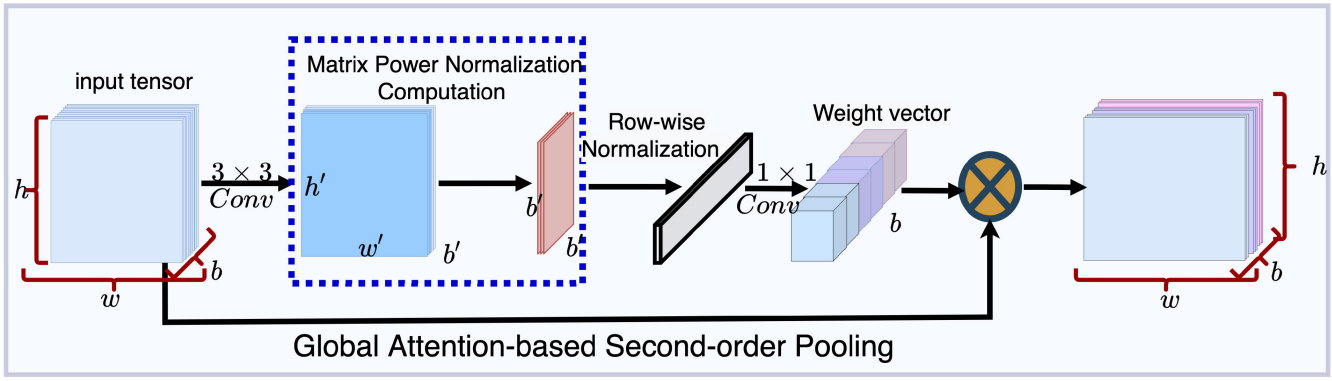


Fig. 4. Schematic illustration of the global spectral–spatial attention-based second-order pooling.

statistical features of HSI according to (14) using the first-order statistical features F_{1st} obtained from initial layers

$$F_{\text{Cov}} = F_{1st}^T F_{1st}. \tag{14}$$

$F_{\text{Cov}} \in R^{N\times N}$ is a symmetric matrix that computes the inner product of each row–column vector pair in F_{1st} to produce more discriminative features. A covariance pooling mechanism aggregates second-order statistical features and captures holistic HSI information. Yet, it cannot model the diversity of neighboring pixels, which may present varying information. We present a covariance pooling module with an enhanced self-attention mechanism that models pairwise correlations of the holistic HSI information to capture the global spectra-spatial second-order statistics, thereby making full use of contextual image information, which is illustrated in Fig. 4. The schematic is composed of a squeeze and an excitation module. We perform covariance normalization by computing the power operation through an eigenvalue decomposition and use the Newton–Schulz iteration to accelerate the decomposition process in order to enable the fast execution and integration of the covariance matrix into the pooling module.

Letting $X \in \mathbb{R}^{B\times H\times W}$ be the input feature of the lth layer of the network with H, W, and B being the spatial height, the width, and the number of spectral bands, respectively, we formulate the covariance normalization as

$$M = \rho(X; \Omega, \beta) \tag{15}$$

where ρ in (15) denotes the CNN feature function with a series of convolutional and pooling layers. Ω and β are the feature function's weight and biases, respectively. The feature map $M \in \mathbb{R}^{B\times H\times W}$ is reshaped into $M \in \mathbb{R}^{d\times n}$, with $n = W \times H$ features with dimensions d. We embed the covariance differentiation into the deep CNN (DCNN) to establish the covariance matrix $\boldsymbol{F}$. The covariance matrix can then be rewritten as

$$\boldsymbol{F} = M\bar{\mathbf{I}}M^T \tag{16}$$

where $\bar{\mathbf{I}} = (1/n)(\mathbf{I}-(1/n)JJ^T)$ is the identity matrix of $n\times n$. J is the d-dimensional vector with all entries being ones, and J^T is the transpose matrix of J. With F being symmetric, we perform the eigenvalue decomposition using (17), which we then use to compute the power operation of the covariance matrix

$$\boldsymbol{F} \to (\mathbf{U}, \Lambda) \tag{17}$$

where $\Lambda = \text{diag}(\lambda_1, \lambda_2, \ldots, \lambda_d)$ is the diagonal matrix with the eigenvalues $\lambda_i = 1, 2, \ldots, d$ being arranged in a nonincreasing order, and $\mathbf{U} = [u_1, u_2, \ldots, u_d]$ is the orthogonal matrix where the column u_i is the corresponding eigenvector for the eingenvalue λ_i. Thereby, to perform the power operation of the covariance matrix via the eigenvalue decomposition, we formulate the decomposition as follows:

$$(\mathbf{U}, \Lambda) \to \mathbf{Z}, \mathbf{Z} = \boldsymbol{F}^\alpha = \mathbf{U}\Theta(\Lambda)\mathbf{U}^T. \tag{18}$$

From (18), $\Theta(\Lambda) = \text{diag}(\lambda_1^\alpha, \lambda_1^\alpha, \ldots, \lambda_d^\alpha)$, with α being a positive real number. In this study, the value of α that leads to higher performance in capturing diverse pixels for improved classification of HSI is 0.5.

2) Accelerating Covariance Pooling Normalization: Since eigenvalue decomposition that the second-order pooling heavily relies on is slow when implemented on a graphical processing unit (GPU) [38], [39], we use the Newton–Schulz iteration to hasten the computational speed of the covariance normalization process [38]. From (18), $Z = \Upsilon = \boldsymbol{F}^\alpha$, which can further be rewritten as $\boldsymbol{F}^\alpha = \mathbf{U}\text{diag}(\lambda_i^\alpha)\mathbf{U}^T$, where $\alpha = 0.5$ in this studies. Letting $\Upsilon_1 = \hat{\boldsymbol{F}}, Z = \mathbf{I}\forall n = 1, \ldots, d$, following the proposition in the work of Li et al. [38], we formulate the couple iteration, as provided in the following equation:

$$\begin{aligned}\Upsilon_n &= \frac{1}{2}\Upsilon_{n-1}(3\mathbf{I} - Z_{n-1}\Upsilon_{n-1}) \\ Z &= \frac{1}{2}(3\mathbf{I} - Z_{n-1}\Upsilon_{n-1})Z_{n-1}.\end{aligned} \tag{19}$$

The formulation in (19) is then computed using matrix product to obtain an approximated solution with less number of iterations. It should be noted that, in this study, the number of iterations is 7

$$\hat{\boldsymbol{F}} = \frac{1}{\text{tr}(\boldsymbol{F})}\boldsymbol{F} \tag{20}$$

where $\text{tr}(\boldsymbol{F})$ is the trace of matrix $\boldsymbol{F}$. Equation 20 is then used to ensure that the Newton–Schulz iteration converges in time.

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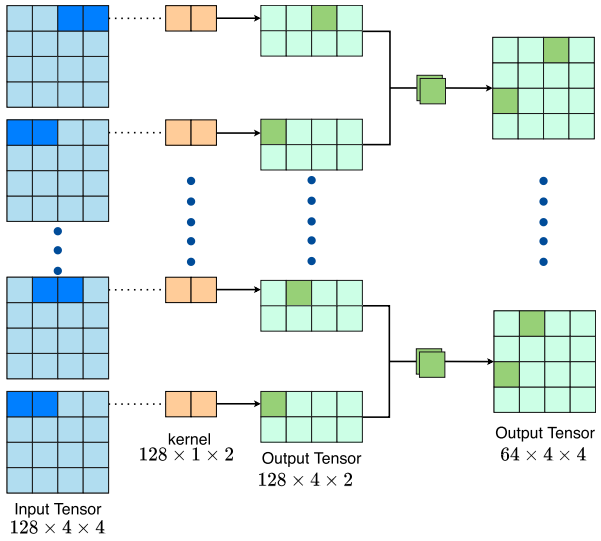


Fig. 5. Implementation of variable pixel extraction, where the spatial information by the horizontal dot product and the spectral features are extracted and fused using feature concatenation.

3) Context Block Implementation Using Covariance Pooling: The squeeze module models the second-order statistics along the spectral dimension. A 1×1 convolution is used to reduce the number of spectral bands to b', where $b' < b$ reduces the dependence of weakened features. We then independently compute the pairwise correlations from the reduced tensor $h' \times w' \times b'$ along the spectral to produce covariance matrix $b' \times b'$, as shown in Fig. 4. Similarly, the spatial dimensions $h' \times w'$ are modeled to obtain the spatial covariance matrix $h'w' \times h'w'$. This is an extension of modeling features not only along the spectral-band dimensions to capture the pairwise feature correlations of the holistic feature maps for spatial-feature scaling, as provided in Fig. 5, suggesting how the i_{th} row of the feature map's covariance matrix, where $i = 1, 2, \ldots, b'$ or $i = 1, 2, \ldots, h'w'$, has statistical correlations with all the feature along both the spectral and spatial dimensions, respectively. We then perform a rowwise normalization operation on the covariance matrices.

In this way, the structure of the covariance matrix is preserved before feeding it to the excitation module. The excitation modules consist of two sequential operations. First, a rowwise convolution is performed for the spectral–spatial covariance matrices in a group convolution. After another convolution and a nonlinear activation in this instance, we used the sigmoid activation function to produce a spectral weight vector $b \times 1$ and a spatial weight matrix $h \times w$ after proper reshaping. The scaling is conducted by multiplying each spectral band element in the input features with the weight vectors and the weight matrix with the spatial input features to suppress nonsignificant features after upsampling $b \times 1$ and $h \times w$. Following the pixel-level features convolution modeling protocol in work done by Liu et al. [29], the 3-D kernel is separated into 2-D and 1-D to extract spectral features and spatial features, respectively. This improves the robustness of the model and avoids overfitting, especially for a small dataset. Letting τ^l be the output feature of the pixel-level

TABLE I
CLASS DISTRIBUTIONS OF THE UNIVERSITY OF INDIANA PINES DATASET

No.	Class	Train	Val	Test
1	Alfalfa	5	1	48
2	Corn-notill	143	14	1277
3	Corn-mintill	83	8	743
4	Corn	23	2	209
5	Grass-Pasture	49	4	444
6	Tree-Grass	74	7	666
7	Grass-pasture-moved	2	1	23
8	Hay-windrowed	48	4	437
9	Oats	2	1	17
10	Soybeans-notill	96	9	863
11	Soybeans-mintill	246	24	2198
12	Soybeans-clean	61	6	547
13	Wheat	21	2	189
14	Woods	129	12	1153
15	Building-Grass-Trees-Drives	38	3	339
16	Stone-steel-Towers	9	1	85
	Total	1029	99	9238

convolution layer, we formulate the convolution operation as

$$\tau_i^l = f\left(\sum_{p_n \in \mathbb{R}} k_j^j(p_n) \cdot f\left(W_j^j \cdot \text{BN}(p_0 + p_1)\right) + \beta_j^j\right) \quad (21)$$

where $\text{BN}(p_0 + p_1)$ denotes the normalized input of the l_th layer, and k_j^j, W_j^j, and β_j^j are the 1-D spectral kernel, the 2-D spatial kernel, and the bias, respectively. P_n lists the positions of the sampling grid $\mathbb{R}$, which, for this work, is given by a 3×3 sampling grid. As shown in Fig. 5, the pixel-level features extracted by the attention model are applied to detect spatial regions. The variable pixel convolution block, as shown in Fig. 5, is designed to integrate the spectral features and the spatial information of the second-order feature maps for feature extraction, where the horizontal convolution is first exploited to extract spatial information, and then, the vertical stitching is utilized to correlate feature spectral information.

Similar to other modules, we let SSAN consist of l_an, where $1 < l_an$, having an input and output to be τ^0 and τ_{an}^l, respectively, thereby giving the pixel-level convolution operation as $\tau_{\text{an}}^l = \text{SSAN}(\tau^0)$.

D. Local Pixel-Level Feature and Superpixel-Level Feature Fusion

The proposed method constitutes three modules: SRN, SGN, and SSAN. Each of these constituting modules is defined to undertake a specific function. The SRN plays a crucial role in our proposed method. Its purpose is to reduce the redundant information in the spectra and enhance their discrimination, which, in turn, provides improved initial features for both the pixel-level SSAN and the superpixel-based GCN. The output of the SRN is sent to the encoding network to enhance the representation learning process and generate improved superpixel-level features. SGN models the representations of HSIs on graphs by generating and extracting relevant superpixel-level features. SSAN utilizes covariance pooling to exploit the second-order statistics of the holistic neighboring pixels from the HSI via a spectra-spatial attention mechanism. We fused representations of all three modules

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TABLE II
CLASS DISTRIBUTIONS OF THE UNIVERSITY OF PAVIA DATASET

No.	Class	Train	Val	Test
1	Asphalt	67	67	6497
2	Meadows	187	187	18275
3	Gravel	21	21	2057
4	Trees	31	31	3002
5	Painted metal sheets	14	14	1317
6	Baresoil	51	51	4927
7	Bitumen	14	14	1302
8	Self-Blocking Bricks	37	37	3608
9	Shadows	10	10	927
	Total	432	432	41912

by formulating the pixel-level and superpixel-level feature representation fusion as

$$F^* = \phi\big(\mathrm{SGN}\big(\Phi\big(\mathbf{X}^{L_{\mathrm{sc}}} : \mathbf{Q}\big); \mathbf{M}\big) : \mathbf{Q}\big) \oplus \mathrm{SSAN}\big(\mathbf{X}^{L_{\mathrm{sc}}}\big). \quad (22)$$

F^* is the fused features that are concatenated $\oplus$ along the spectral dimensions to produce the fused feature representation map from the superpixel-based graph features and the spectral–spatial pixel-level features. From (22), Φ encodes the enhanced HSI (X_{sc}^L) that is obtained from SRN and the association matrix Q to produce the feature vector V_n of the nth node in the feature matrix V when the SGN is executed. ϕ decodes by restoring and propagating the encoded graph node feature to pixel-level features together with Q following the general formulation given in (6). Practically, $\phi = \mathrm{Decoder}(V; Q) = \mathrm{Reshape}(QV)$. We use a softmax classifier to compute the probability scores for each class to calculate at each pixel. Letting Γ be the output probability map and then for each position of a pixel $\mathbf{P}_0$ in Γ. We compute $\Gamma_{(\mathbf{P}_0)}$ by utilizing the formulation in the following equation:

$$\Gamma_{(\mathbf{P}_0)} = \frac{1}{\sum_n^L e^{k_n \cdot \boldsymbol{F}^*(\mathbf{P}_0)+\beta_n}} \begin{bmatrix} e^{k_1 \cdot \boldsymbol{F}^*(\mathbf{P}_0)+\beta_1} \\ e^{k_2 \cdot \boldsymbol{F}^*(\mathbf{P}_0)+\beta_2} \\ \vdots \\ e^{k_N \cdot \boldsymbol{F}^*(\mathbf{P}_0)+\beta_N} \end{bmatrix}. \quad (23)$$

$\beta_n (1 \le n \le N)$ and $\mathbf{k}_n (1 \le n \le N)$ from (23) represent the n_th bias and 1-D kernel, respectively. $\Gamma_{(\mathbf{P}_0)}$ is the probability vector of each class. We train the network using the cross-entropy loss function as provided in the following equation:

$$\mathrm{CE}(\mathrm{p}_t) = -\sum_{t=1}^{N}\sum_{v=1}^{V} \mathbf{L}_c(\mathbf{p}_t) \log\big(\Gamma_c(\mathbf{p}_t)\big). \quad (24)$$

From (24), $\mathbf{L}_v(\mathbf{p}_t)$ represents the c_th the element of the label, $\Gamma_v(\mathbf{p}_t)$ denotes the pixel probability in a HSI, and $\mathbf{X}(\mathbf{p}_t)$ denotes that it belongs to the c_th class.

III. EXPERIMENTATION

The PiCovS consists of three modules: SRN, SSAN, and SGN, and the effectiveness of the proposed PiCovS for HSI classification is evaluated across four hyperspectral benchmark datasets with details provided in Tables I–IV. The performance results obtained are then compared to six state-of-the-art deep learning methods for HSI classification: DCNN [49], spectral–spatial residual network (SSRN) [50], fast dense spectral–spatial convolutional network (FDSSC) [16], hybrid spectral network (HybridSN) [51], double-branch dual-attention network (DBDA) [52], and CEGCN [29]. We use the same experimental settings, including the hyperparameters recommended for all the baseline approaches provided in their respective original works. Furthermore, the overall accuracy (OA), average accuracy (AA), and kappa (KPP) statistics were employed as the evaluation metrics to evaluate the performance of the proposed model. OA is defined as the ratio of correctly classified samples among the total test samples. AA is the average of classwise accuracy, and KPP measures strong mutual agreement between the given ground-truth map and the classification map generated by the model.

TABLE III
CLASS DISTRIBUTIONS OF THE SALINAS DATASET

No.	Class	Train	Val	Test
1	Brocoli_green_weeds_1	21	21	1967
2	Brocoli_green_weeds_2	38	38	3650
3	Fallow	20	20	1936
4	Fallow_rough_plow	14	14	1366
5	Fallow_smooth	27	27	2624
6	Stubble	40	40	3879
7	Celery	36	36	3507
8	Grapes_untrained	113	113	11045
9	Soil_vinyard_develop	63	63	6077
10	Corn_senesced_green_weeds	33	33	3212
11	Lettuce_romaine_4wk	11	11	1046
12	Lettuce_romaine_5wk	20	20	1887
13	Lettuce_romaine_6wk	10	10	896
14	Lettuce_romaine_7wk	11	11	1048
15	Vinyard_untrained	73	73	7122
16	Vinyard_vertical_trellis	19	19	1769
	Total	549	549	53031

TABLE IV
CLASS DISTRIBUTIONS OF THE UNIVERSITY OF HOUSTON 2018 DATASET

No	Class	Train	Val	Total
1	Healthy grass	1458	8341	9799
2	Stressed grass	4316	28186	32502
3	Artificial Turf	331	353	684
4	Evergreen trees	2005	11583	13588
5	Deciduous trees	676	4372	5048
6	Bare earth	1757	2759	4516
7	Water	147	119	266
8	Residential buildings	3809	35953	39762
9	Non-residential buildings	2789	220895	223684
10	Roads	3188	42622	45810
11	Sidewalks	2699	31303	34002
12	Crosswalks	225	1291	1516
13	Major thoroughfares	5193	41165	46358
14	Highways	700	9149	9849
15	Railways	1224	5713	6937
16	Paved parking lots	1179		11475
17	Unpaved parking lots	127	22	149
18	Cars	848	5730	6578
19	Trains	493	4872	5365
20	Stadium seats	1313	5511	6824
	Total	34477	470235	504712

A. Experimental Settings

Experiment were performed using segmentation scale values λ of 100, 200, 300, 400 and 500 in the SGN to generate the superpixels and their influence on the proposed PiCovS

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TABLE V
ARCHITECTURE SETTINGS OF THE PICOVS. λ IS THE SUPERPIXEL SCALE

Module	Layer No.	Size and Number of Kernels
SRN	1	1×1@128
	2	1×1@128
SSAN	1	3 ×3@128
	2	3×3@8
SGN	1	@128
	2	@64
	λ = 100	

performance. It should be noted that, although various segmentation scales were tested, the results reported in subsequent sections are obtained by experimenting with $\lambda = 100$. Similarly, the depth of layers for the three modules, SRN, SSAN, and SGN, were experimented with utilizing variations of layer depths; hence, $L_{SR} = L_{SS} = L_{SG} = [1, 2, 3, 4]$. The SSAN's unique convolutional kernel size is set to 3×3. PiCovS's activation functions are all leaky-rectified linear units (Leaky-ReLUs) [54]. Table V summarizes other specific settings. Finally, we train our network for 100 epochs and use the Adam optimizer [55] to optimize the cross-entropy loss function in (24) with a learning rate of $5e-4$. All models are implemented with Python-3.6 using the PyTorch framework and run on a computer furnished with an AMD Ryzen 9 5900x CPU, 32 RAM, and a Nvidia GTX-3060 GPU.

B. Dataset

We evaluated the proposed PiCovS on four hyperspectral benchmarks: Indian Pines, University of Pavia, Salinas, and Houston University-2018 datasets. Description details and class distribution of each benchmark are provided in the following and in Tables I–IV for Indian Pines, Pavia University, Salinas, and Houston University-2018 datasets, respectively.

1) Indian Pines: The Indian pines data is the first dataset on which we evaluate PiCovS. It was collected over the agricultural test location in Northwest Indiana using an Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor. This HSI consists of 220 bands with a pixel size of 145 × 145. Its landscape is primarily agricultural, with the remaining third being forest or other types of perennial natural flora. After reducing the number of bands by removing the region covered with water vapor absorption bands, the remaining 200 bands making up of 10 366 samples from 16 terrain classes were chosen for the study and analysis. In addition, a random selection process of 10%, 1%, and 89% of samples per class for training, validation, and testing, respectively, was conducted with the corresponding values presented in Table I.

2) University of Pavia: The University of Pavia provided the second benchmark dataset for evaluating the proposed method. The ROSIS sensor gathered it throughout the flying campaign over Pavia, Northern Italy. With 115 bands and 610 × 340 pixels, the University of Pavia dataset has a geometric resolution of 1.3 m. Bands without information on image samples were suppressed and not utilized during model training. Therefore, 103 bands were used to evaluate the proposed method. Nine different terrain classifications making a total of 42 776 samples were finally used to evaluate the proposed model. The dataset was split into 1%, 1%, and 98% samples for training, validation, and testing per class. The distribution summary is presented in Table II.

3) Salinas: The third dataset, the Salinas scene, was collected over the Salinas Valley in California and had an HSR of 3.7-m pixels using an AVIRIS sensor. It comprises 224 bands that are 512 × 217 in size. We reduced the number of bands by removing the region covered by water vapor absorption with the remaining 204 bands being used to train the model, thereby using 54 129 samples from 16 crops for training and testing the model. Then, as training, validation, and testing datasets, 1%, 1%, and 98% of samples per class are chosen randomly. Table III shows the complete class distribution information of the Salinas datasets and the train–test split.

4) Houston 2018: The Houston 2018 dataset [56] covers the University of Houston area and was acquired on February 16, 2017, using the Compact Airborne Spectrographic Imager (CASI) 1500. This dataset comprises 48 spectral bands, ranging from 0.38 to 1.05 μm. The image has a ground sampling distance of 1.3 m. For our experiments, we utilize the training portion of the dataset, which was made available by the University of Houston and the Image Analysis and Data Fusion Technical Committee for the data fusion contest. The dataset includes diverse features, such as complex buildings, roads, and other areas, categorized into 20 ground-truth classes. The image size is 601 × 2384 pixels, and a total of 504 712 samples are considered for evaluation, as indicated in Table IV.

C. Comparison of Classification Performance

Tables VI–IX provide the mean and standard deviation values for OA, AA, and KPP coefficient metrics based on ten experimental evaluation of the proposed PiCovS and other CNN-based state-of-the-art methods, as well as each evaluation index for superpixel-based hybrid methods that the proposed method was compared to together with classification maps produced by these techniques, as shown in Figs. 6–9, for the Indian Pines, Salinas, University of Pavia, and the Houston University-2018 datasets, respectively. The strategy of running the experiments ten times prevents the impact of weight initialization and randomness of data sampling. It could be seen from the tables, on the whole, that the proposed method obtained superior OA, AA, and KPP values demonstrating the effectiveness and efficiency of the proposed method when it was compared to all the other methods.

On the Indian pines dataset, our method obtains an OA value of 99.84%, and the CNN-based state-of-the-art methods, DCNN, SSRN, FDSSC, HybridSN, and DBDA, achieved 96.31%, 97.86%, 98.93%, 98.44%, and 98.24%, respectively. On the other hand, the proposed PiCovS obtained the best classification results. Compared to other hybrid state-of-the-art methods, the proposed method (PiCovS) outperformed the CNN-GCN hybrid networks 96.86 ± 0.37, 94.3 ± 2.84, 98.27 ± 0.28, 99.12 ± 0.19, and 99.84 ± 0.0 with a margin of 2%, 4%, 1%, and 0.72% for WFCG, EMS-GCN, FDGC, and CEGCN methods, respectively. The performance rate obtained by the proposed PiCovS could be attributed to the use of the SSAN to capture pixel-level features and introduce diversity among neighboring

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TABLE VI

INDIVIDUAL CLASS, OA, AA, AND KPP OF ALL MODELS ON THE INDIAN PINES DATASET

Method/Class	DCNN[50]	SSRN[51]	FDSSC[16]	HybridSN[52]	DBDA[53]	WFCG [10]	EMS-GCN[31]	FDGC[54]	CEGCN[30]	PiCovS(Ours)
1	98.12±3.35	99.42±1.14	**100**±0	97.16±2.73	**100**±0	91.57±2.45	97.5±3.06	85.4±9.87	97.50±2.04	97.79±0.34
2	94.72±1.69	98.40±0.96	98.29±0.62	97.15±0.56	97.79±1.05	90.52±1.16	91.21±0.59	97.85±1.1	99.08±0.57	**99.74**±0.11
3	94.33±2.30	97.13±2.29	98.19±0.78	98.25±0.65	98.56±1.08	85.59±3.20	97.13±1.7	99.59±0.35	99.27±0.41	**99.80**±0.01
4	94.32±4.47	96.71±4.20	98.30±1.12	97.90±2.57	94.89±1.49	79.53±6.70	96.71±1.91	97.42±0.54	97.98±1.46	**99.83**±0.13
5	98.08±0.92	97.20±1.30	98.06±0.17	98.49±0.96	97.54±1.44	98.02±0.29	92.53±3.30	98.86±0.6	99.14±0.88	**99.83**±1.17
6	98.79±0.74	99.01±0.89	98.98±0.57	98.92±0.38	98.39±1.57	94.01±1.54	98.43±0.89	94.71±0.57	99.63±0.32	**100**±0
7	**100** ±0	80±40	**100**±0	**100**±0	91.81±16.36	58.89 ±7.95	96.64±0.18	**100**±0	99.13±1.73	98.46±0.81
8	97.75±1.64	99.45±0.61	99.95±0.09	99.67±0.31	99.64±0.71	95.71±4.41	98.45±2.19	99.38±1.23	99.90±0.11	**100**±0
9	90 ±30	0±0	60±48.98	92.38±5.25	**98.75**±2.50	73.91±4.73	**100**±0	92.5±15.0	97.77±4.44	98.8±0.54
10	95.53 ±2.45	96.38±3.06	98.72±1.08	98.74±0.81	96.38±2.01	92.27 ±7.55	91.38±4.26	97.65±1.08	97.03±1.21	**99.83**±0.06
11	96.66 ±1.18	98.02±0.90	99.34±0.36	99.16±0.26	98.87±0.32	92.35 ±6.38	94.02±0.88	99.58±0.19	99.49±0.17	**99.82**±0.15
12	94.87 ±3.02	95.42±4.10	98.19±0.88	97.47±1.17	98.67±0.95	81.51 ±0.34	96.92±1.40	97.0±2.96	99.26±0.25	**99.56**±0.41
13	99.37 ±0.69	98.95±1.27	**100**±0	98.02±1.64	99.37±0.76	93.15 ±2.86	99.51±0.27	96.79±3.24	97.43±1.08	**100**±0
14	98.36 ±0.90	99.48±0.28	99.75±0.28	99.32±0.44	99.66±0.27	97.50 ±1.92	97.81±0.30	99.11±0.4	**100**±0	99.96±0.1
15	93.94 ±1.85	96.76±1.89	98.88±0.75	97.64±1.58	97.82±1.74	86.23 ±5.80	**99.83**±1.89	96.91±1.62	99.22±0.78	**99.79**±0
16	92.43 ±5.34	97.05±2.63	98.37±2.12	91.02±4.07	89.54±4.91	95.37 ±3.34	98.05±3.41	94.11±6.18	96.68±3.21	**99.68**±0.02
OA (%)	96.31±0.37	97.86±0.41	98.93±0.27	98.44±0.16	98.24±0.35	96.86±0.37	94.31±2.84	98.27±0.28	99.12±0.19	**99.84**±0.06
AA(%)	96.08±2.10	90.59±2.34	96.56±2.96	97.58±0.82	97.35±1.12	87.88±2.10	95.73±3.33	96.81±1.02	98.66±0.37	**99.42**±0.04
KPP(×100)	95.80±0.43	97.56±0.47	98.78±0.31	98.23±0.18	97.99±0.40	89.58±0.43	94.01±1.19	98±0.33	99.01±0.22	**99.78**±0.21

NB: Boldface indicates the highest value obtained.

TABLE VII

INDIVIDUAL CLASS, OA, AA, AND KPP OF ALL MODELS ON THE UNIVERSITY OF PAVIA DATASET

Method/Class	DCNN[50]	SSRN[51]	FDSSC[16]	HybridSN[52]	DBDA[53]	WFCG [10]	EMS-GCN[31]	FDGC[54]	CEGCN[30]	PiCovS(Ours)
1	92.72±1.02	99.66±0.24	99.22±0.67	95.13±1.81	97.48±1.92	91.67±5.56	98.11±0.16	85.4±9.87	98.44±2.64	**99.98±0**
2	97.14±0.59	98.70±1.02	99.62±0.27	99.16±0.49	99.36±0.61	96.52±0.06	89.83±2.64	97.85±1.1	**100**±0	99.97±0.40
3	87.91±3.40	93.95±5.15	97.54±1.96	88.73±4.90	99.19±1.23	81.91±2.90	99.05±1.72	99.59±0.35	99.79±0.19	**99.90±0.04**
4	99.35±0.90	99.72±0.27	**99.89±0.09**	98.18±0.77	98.12±0.29	92±1.01	96.35±0.38	97.42±0.54	96.35±0.73	99.84±1.01
5	98.92±1.61	99.93±0.08	99.29±1.22	98.98±0.93	99.48±0.51	94.13±1.74	**100**±0	98.86±0.22	100±0.88	**100±0**
6	97.41±0.77	98.52±2.11	99.56±0.48	98.66±0.96	98.86±0.90	97.14±1.41	97.76±0.91	94.71±0.57	99.85±0.15	**100±0**
7	91.99±4.80	96.84±2.22	98.52±2.25	96.64±2.37	97.18±5.46	86.89±1.21	98.90±0.08	**100**±0	99.98±0.03	**100±0**
8	88.41±1.07	88.85±5.74	94.29±2.27	90.69±2.72	88.86±5.60	85.45±2.31	99.45±2.19	99.38±1.23	99.46±0.49	**99.99±0.01**
9	99.41±0.59	99.53±0.55	99.72±0.23	97.21±1.86	97.56±2.88	98.36±3.37	**100**±0	92.5±15.0	99.87±0.08	99.13±1.09
OA (%)	95.35±0.17	97.54±0.58	98.94±0.21	97.01±0.69	97.74±0.79	93.56±0.73	98.13±0.27	98.27±0.28	97.12±1.01	**99.97±0.01**
AA(%)	94.81±0.56	97.30±0.55	98.63±0.47	95.93±0.87	97.34±0.93	90.29±3.12	98.81±0.15	96.81±1.02	98.66±0.07	**99.95±0.02**
KPP(× 100)	93.81±0.23	96.74±0.77	98.60±0.28	96.02±0.92	97.01±1.05	91.33±3.03	96.99±0.19	98±0.33	97.15±0.82	**99.91±2.05**

NB: Boldface indicates the highest value obtained.

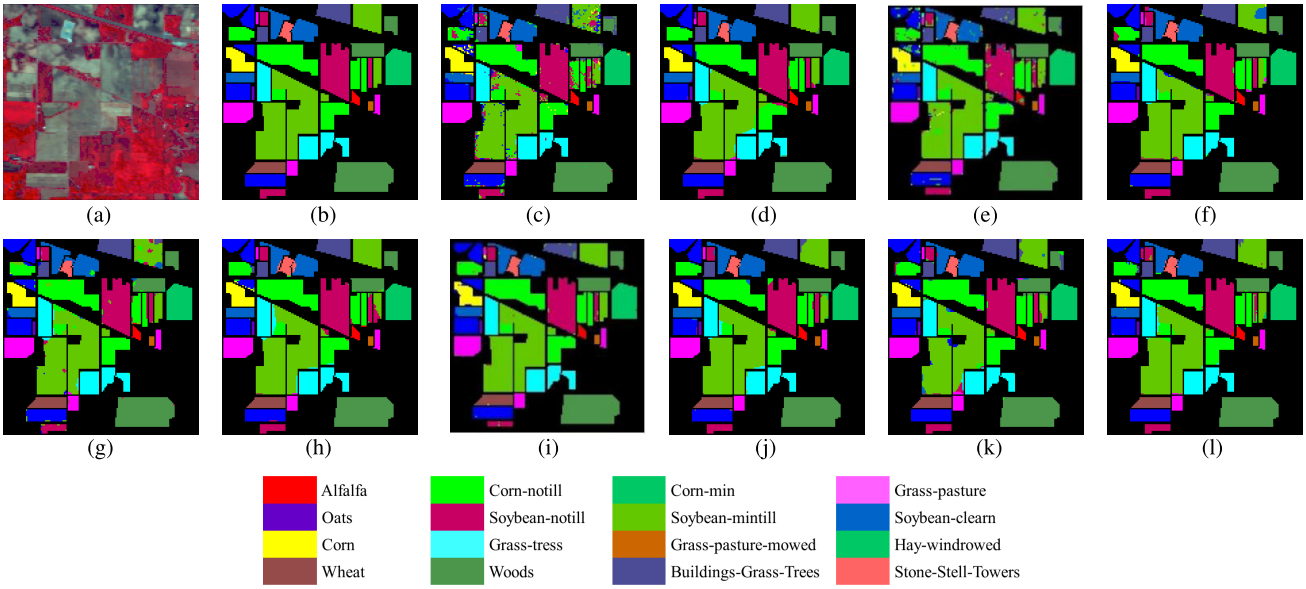


Fig. 6. Classification maps on the Indian Pines dataset from various state-of-the-art methods. (a) False color image. (b) Ground truth. (c) DCNN (96.31%). (d) SSRN (97.86%). (e) FDSSC (98.93%). (f) HybridSN (98.44%). (g) DBDA (98.24%). (h) WFCG (96.86%). (i) EMS-GCN (94.13%). (j) FDGC (98.27%). (k) CEGCN (99.12%). (l) PiCovS (99.84%).

pixels to extract features along the spectral–spatial dimensions and the modified manifold SLIC to incorporate texture information for the superpixel generation for the SGN module to capture the large-scale contextual features of HSI and

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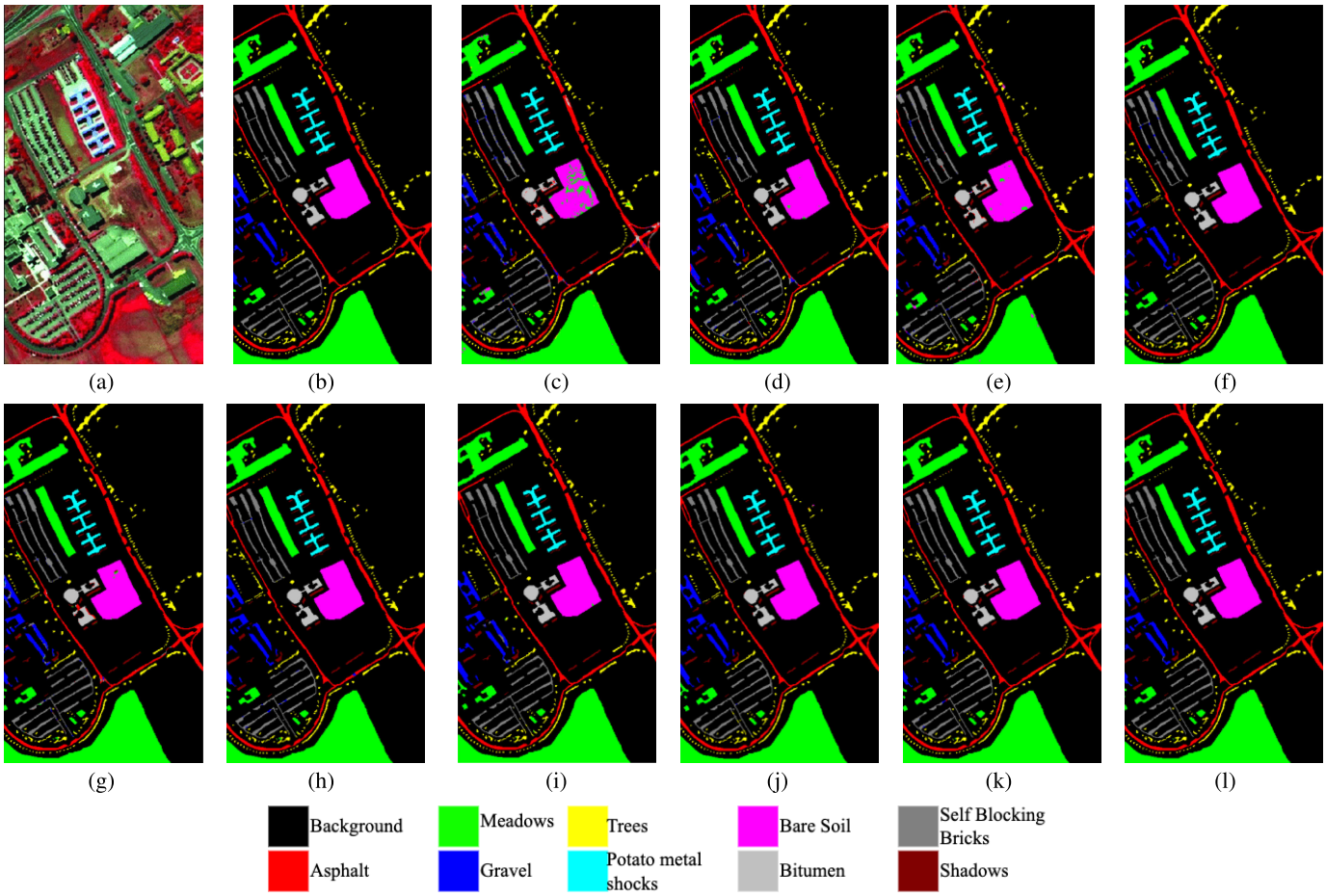


Fig. 7. Classification maps on the University of Pavia dataset from various state-of-the-art methods. (a) False color image. (b) Ground truth. (c) DCNN (95.35%). (d) SSRN (97.54%). (e) FDSSC (98.94%). (f) HybridSN (97.01%). (g) DBDA (97.74%). (h) WFCG (93.56%). (i) EMS-GCN (98.13%). (j) FDGC (98.27%). (k) CEGCN (97.12%). (l) PiCovS (99.97%).

TABLE VIII

INDIVIDUAL CLASS, OA, AA, AND KPP OF ALL MODELS ON THE SALINAS DATASET

Method/Class	DCNN[50]	SSRN[51]	FDSSC[16]	HybridSN[52]	DBDA[53]	WFCG [10]	EMS-GCN[31]	FDGC[54]	CEGCN[30]	PiCovS
1	98.12±3.35	99.42±1.14	**100**±0	97.16±2.73	**100**±0	91.57±2.45	97.5±3.06	99.31±9.87	**100±0**	97.79±0.34
2	94.72±1.69	98.40±0.96	98.29±0.62	97.15±0.56	97.79±1.05	90.52±1.16	91.21±0.59	**100**±0	**100±0**	99.74±0.11
3	94.33±2.30	97.13±2.29	98.19±0.78	98.25±0.65	98.56±1.08	85.59±3.20	97.13±1.7	97.92±0.35	99.27±0.41	**99.80**±0.01
4	94.32±4.47	96.71±4.20	98.30±1.12	97.90±2.57	94.89±1.49	79.53±6.70	96.71±1.91	98.71±0.54	99.86±0.19	**99.83**±0.13
5	98.08±0.92	97.20±1.30	98.06±0.17	98.49±0.96	97.54±1.44	98.02±0.29	92.53±3.30	99.0±0.06	98.65±0.84	**99.83**±1.17
6	98.79±0.74	99.01±0.89	98.98±0.57	98.92±0.38	98.39±1.57	94.01±1.54	98.43±0.89	99.71±0.12	99.63±0.32	**100**±0
7	**100** ±0	80±40	**100**±0	**100**±0	91.81±16.36	58.89 ±7.95	96.64±0.18	**100**±0	99.97±0.02	98.46±0.81
8	97.75±1.64	99.45±0.61	99.95±0.09	99.67±0.31	99.64±0.71	95.71±4.41	98.45±2.19	98.38±1.02	99.90±0.11	**100**±0
9	90 ±30	0±0	60±48.98	92.38±5.25	**98.75**±2.50	73.91±4.73	**100**±0	99.25±1.80	**100±0**	98.8±0.54
10	95.53 ±2.45	96.38±3.06	98.72±1.08	98.74±0.81	96.38±2.01	92.27 ±7.55	91.38±4.26	98.19±1.08	97.99±1.70	**99.83**±0.06
11	96.66 ±1.18	98.02±0.90	99.34±0.36	99.16±0.26	98.87±0.32	92.35 ±6.38	94.02±0.88	99.81±0.19	99.49±0.17	**99.82**±0.15
12	94.87 ±3.02	95.42±4.10	98.19±0.88	97.47±1.17	98.67±0.95	81.51 ±0.34	96.92±1.40	99.01±3.01	98.66±0.25	**99.56**±0.41
13	99.37 ±0.69	98.95±1.27	**100**±0	98.02±1.64	99.37±0.76	93.15 ±2.86	99.51±0.27	99.63±0.4	99.77±0.05	**100**±0
14	98.36 ±0.90	99.48±0.28	99.75±0.28	99.32±0.44	99.66±0.27	97.50 ±1.92	97.81±0.30	99.07±0.74	98.87±1.32	99.96±0.1
15	93.94 ±1.85	96.76±1.89	98.88±0.75	97.64±1.58	97.82±1.74	86.23 ±5.80	**99.83**±1.89	98.97±1.28	99.22±0.02	99.79±0
16	92.43 ±5.34	97.05±2.63	98.37±2.12	91.02±4.07	89.54±4.91	95.37 ±3.34	98.05±3.41	94.11±6.18	98.91±1.27	**99.68**±0.02
OA (%)	96.31±0.37	97.86±0.41	98.93±0.27	98.44±0.16	98.24±0.35	90.86±0.37	94.31±2.84	98.78±0.81	98.36±0.41	**99.84**±0.06
AA(%)	96.08±2.10	90.59±2.34	96.56±2.96	97.58±0.82	97.35±1.12	87.88±2.10	95.73±3.33	98.81±1.09	98.79±0.11	**99.42**±0.04
KPP(×100)	95.80±0.43	97.56±0.47	98.78±0.31	98.23±0.18	97.99±0.40	89.58±0.43	97.01±1.19	98.91±1.08	98.91±0.46	**99.78**±0.21

NB: Boldface indicates the highest value obtained.

model the irregular large-scale spatial interdependent features as well.

On the University of Pavia dataset, our method obtained the highest OA value of 99.97% outperforming the FDSSC (98.94%), the best CNN-based method by a margin of 1% and FDGC (98.27%), and the best performing CNN-GCN method by 1%, further emphasizing the superiority of the proposed model especially when trained with a small amount

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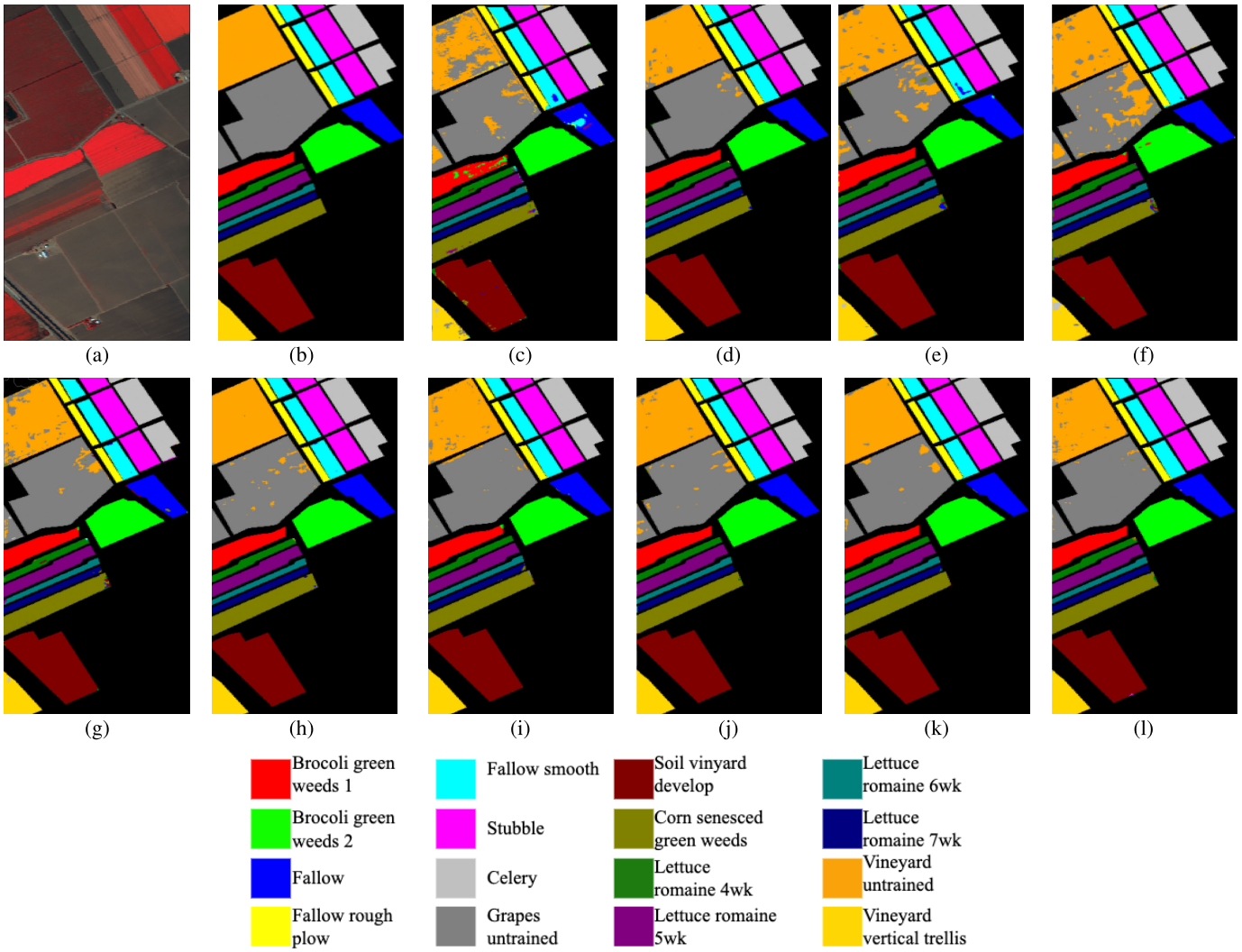


Fig. 8. Classification maps on the Salinas dataset from various state-of-the-art methods. (a) False color image. (b) Ground truth. (c) DCNN (96.31%). (d) SSRN (97.86%). (e) FDSSC (98.93%). (f) HybridSN (98.44%). (g) DBDA (98.24%). (h) WFCG (90.86%). (i) EMS-GCN (94.31%). (J) FDGC (98.78%). (k) CEGCN (98.36%). (l) PiCovS (99.48%).

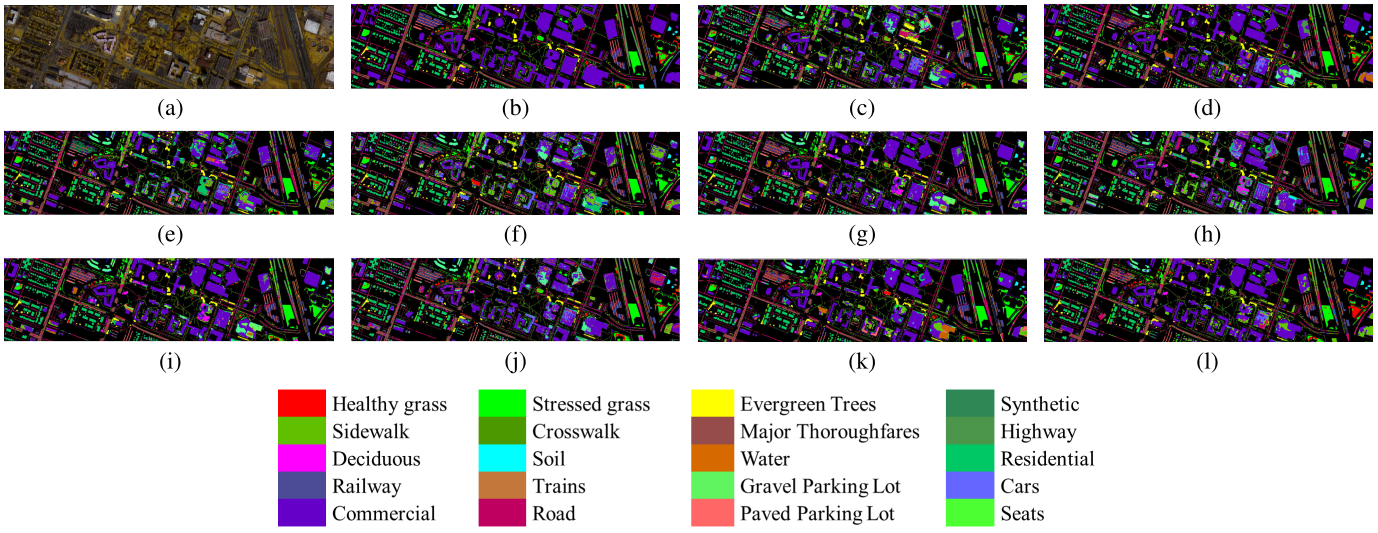


Fig. 9. Classification maps on the University of Pavia dataset from various state-of-the-art methods. (a) False color image. (b) Ground truth. (c) DCNN (55.93%). (d) SSRN (69.16%). (e) FDSSC (69.74%). (f) HybridSN (62.21%). (g) DBDA (56.97%). (h) WFCG (68.05%). (i) EMS-GCN (78.53%). (j) FDGC (70.82%). (k) CEGCN (78.46%). (l) PiCovS (81.96%).

of training data samples, as shown in Fig. 10. It could be observed that the superpixel-based hybrid methods are able to obtain higher OA, A, and KPP coefficients on all four datasets when trained with a limited amount of samples, mainly due

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TABLE IX
INDIVIDUAL CLASS, OA, AA, AND KPP OF ALL MODELS ON THE HOUSTON UNIVERSITY DATASET

Method/Class	DCNN[50]	SSRN[51]	FDSSC[16]	HybridSN[52]	DBDA[53]	WFCG [10]	EMS-GCN[31]	FDGC[54]	CEGCN[30]	PiCovS
1	41.91±4.36	44.43±5.79	41.31±7.16	46.45±8.71	46.85±6.86	52.78±0.15	68.91±5.63	56.66±5.87	**69.26±4.19**	67.39±4.51
2	55.42±8.02	67.83±6.33	67.12±3.43	63.62±5.56	67.31±7.48	74.36±1.27	78.67±2.18	77.89±1.19	79.53±2.76	**81.72±1.62**
3	13.03±14.63	53.61±21.76	43.68±30.18	6.07±1.43	38.53±6.43	85.59±3.12	87.13±11.07	86.14±9.01	88.56±7.41	**89.21±2.38**
4	74.19±2.61	78.81±3.83	77.27±8.15	75.49±8.77	72.69±2.38	79.53±6.70	81.38±2.33	82.57±7.81	**83.29±1.22**	78.81±1.93
5	56.24±9.36	67.52±13.83	54.51±9.69	68.18±2.31	61.17±6.71	59.66±2.69	65.23±0.10	67.84±1.39	58.81±4.46	**69.11±8.37**
6	43.06±6.74	57.90±8.14	57.62±7.42	76.84±4.94	73.18±13.70	71.84±0.47	74.92±5.18	**77.82±9.72**	74.08±6.32	77.19±4.22
7	23.84±015.32	6.55±18.11	10.82±28.37	2.91±9.21	31.73±21.36	2.76±17.54	12.06±16.8	9.89±6.47	11.82±5.96	**61.49±12.80**
8	45.75±5.64	66.71±2.16	67.73±1.09	55.79±2.44	41.15±3.19	65.92±1.71	70.01±0.82	**72.99±6.53**	69.84±1.15	69.22±3.26
9	90.71±4.30	92.88±1.63	92.19±2.47	92.46±1.16	91.76±6.58	88.75±0.18	91.73±0.21	**94.25±1.27**	90.12±3.87	90.78±3.81
10	46.29±7.53	48.64±3.26	57.21±8.34	50.55±1.99	45.12±3.65	41.37 ±4.05	59.34±1.76	44.87±2.16	39.61±1.59	**64.96±2.13**
11	24.13±5.18	33.43±11.1	**36.76±4.75**	24.97±5.29	26.73±5.32	18.58 ±3.16	28.77±1.61	30.01±0.58	29.51±1.74	21.83±7.58
12	8.94±13.01	6.91±4.81	1.67±4.95	3.01±9.02	14.94±5.52	6.01 ±5.42	17.65±1.32	6.11±7.63	11.26±3.25	**18.90±1.63**
13	57.89±7.69	68.29±1.12	65.52±3.29	64.83±1.45	56.96±1.66	69.14 ±1.67	81.97±2.93	70.51±9.12	82.41±6.75	**84.37±9.25**
14	49.09±3.27	56.36±9.01	53.18±9.28	49.01±9.28	54.04±6.29	59.25 ±1.82	63.52±1.09	59.78±3.74	**65.02±11.32**	64.18±2.19
15	74.51±5.38	67.85±16.71	76.42±6.57	87.36±6.39	58.74±8.64	61.53±1.29	85.79±1.89	84.62±3.57	86.37±2.61	**89.38±11.31**
16	64.87±5.34	74.15±2.37	68.63±3.76	68.79±4.53	65.87±4.91	77.41 ±1.93	81.06±0.44	73.83±6.19	80.91±1.74	**82.63±7.59**
17	1.67 ±1.91	12.94±25.83	11.49±16.39	1.78±2.68	10.59±2.68	**82.61±4.76**	76.18±2.16	14.30±8.46	74.67±7.42	78.76±4.67
18	25.33±1.45	39.10±17.82	34.98±7.68	34.59±4.83	29.34±5.83	69.38±6.34	58.31±0.30	40.96±4.88	56.84±4.32	**69.84±14.05**
19	46.89 ±6.82	83.19±5.46	80.24±8.83	72.67±9.47	73.18±3.47	88.89±2.30	87.46±0.16	70.01±8.25	88.19±1.14	**89.39±6.06**
20	36.33 ±5.34	79.07±9.22	49.31±9.47	44.19±20.11	34.64±15.16	64.87 ±1.66	75.62±1.41	49.37±0.11	76.03±1.76	**76.42±2.18**
OA (%)	55.93±1.71	69.16±1.92	69.74±3.71	62.21±5.23	56.97±1.86	68.05±1.19	78.53±1.74	70.82±2.13	78.46±0.51	**81.96±6.04**
AA(%)	44.38±1.82	57.21±2.54	52.85±2.01	49.93±0.74	51.11±1.47	62.14±0.88	76.21±1.07	59.22±2.63	76.10±2.11	**78.36±5.88**
KPP(×100)	48.21±1.79	60.17±3.14	61.22±1.28	51.86±1.06	48.37±6.06	57.36±0.43	65.81±0.35	61.55±0.07	66.93±1.04	**67.14±8.19**

NB: Boldface indicates the highest value obtained.

to the use of pixel-based CNN and superpixel-based GCN to learn and preserve both regular local and irregular high-level information. In contrast to the results obtained on the Indian Pines, University of Pavia, and Salinas datasets, all methods obtained lower OA values when evaluated on the Houston University-2018 dataset, as provided in Table IX. This observation can be attributed to the small appearance of objects and the challenging nature of the dataset. However, the proposed method obtained high values among all comparative methods in regard to OA and KPP coefficient indicating the robustness of introducing the global-attention second-order context block from lower to upper level layers. The qualitative results provided in Figs. 7 and 9 for the University of Pavia and Houston University-2018 datasets, respectively, show a high degree of coherence with the ground truth. The proposed PiCovS ensures smoothed classification outcomes in neighboring regions while keeping features in tiny regions due to how the SSAN and SGN complementarily function. Similarly, Table VIII demonstrates that the proposed PiCovS outperformed all other examined approaches in the experiment using the Salinas datasets.

The classification maps in Figs. 6–9 are provided for the purpose of visual comparison. In general, from the classification maps of comparative methods, the DCNN suffers so much noise in the classification maps for all four datasets. This problem is slightly alleviated by the other CNN-based methods to generate smooth classification maps. The superpixel-based hybrid methods on the other hand relatively obtained smoother classification maps because of their ability to preserve more edge details of classes present in the classification maps although there were few misclassifications. PiCovS produced improved classification maps that are more consistent with the ground truth than other comparative methods. Overall, the proposed PiCovS consistently outperforms the other comparative methods on all four datasets since it is able to suppress the misclassification and noisy scattered points in classification maps mainly due to the global-attention second-order context block to capture long-range features introduce diversity at the pixel-level but also preserves the detailed information well via the use of superpixel-based GCN.

In real-world applications, obtaining well-labeled data is difficult, time-consuming, and expensive. Hence, building a classification method should be data efficient, where the model can be trained with a small number of samples. Fig. 10 shows the impact of the number of training samples on PiCovS and comparable methods. When a large number of training samples is used, PiCovS achieved similar OA on all datasets. PiCovS method's robustness can be clearly observed when PiCovS was trained with five training samples for each class.

Various network depths are evaluated for all datasets, and the OA performances achieved are shown in Fig. 11. Albeit good OA was achieved for all depths, the best OA for almost all the datasets was when the layers of the SSAN and the SGN were all set two. Compared to the University of Pavia dataset, the results obtained for the Salinas in Fig. 11 shows that training with increased depth of the network layers or extending the network branches leads to obtaining high accuracy for some datasets, whereas some datasets would need shallow networks for good OA to be obtained. The stable accuracy values obtained for each evaluation index OA, AA, and KPP for all four datasets suggest that integrating the pixel-level higher order statistical representation and the superpixel-level large-scale representation into a single model leads to achieving robustness and higher efficiency.

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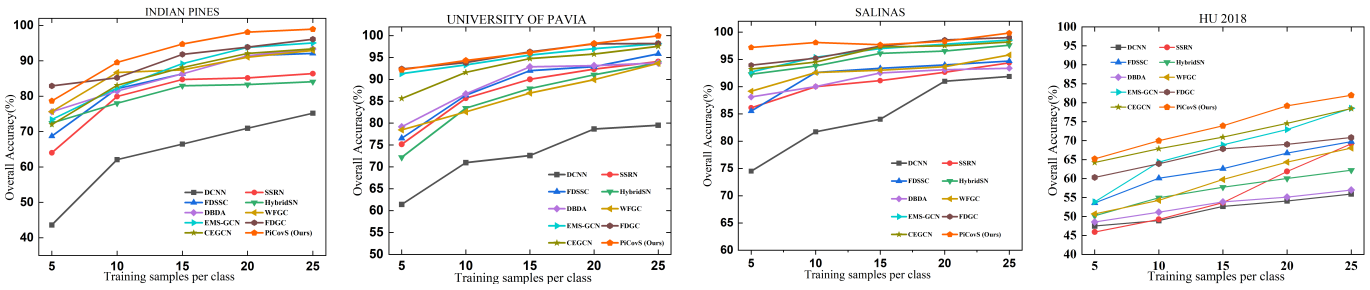


Fig. 10. Comparison of classification of the proposed method to other state-of-the-art methods on various training samples per class. (a) Indian Pines. (b) Pavia University. (c) Salinas. (d) Houston Univesity-2018.

TABLE X
OA (%) INDEXES OF VARIOUS MODULE COMBINATIONS

Dataset	SGN	SSAN	SSAN+SRN	SGN+SRN	SGN+SSAN	SGN+SRN+SSAN
Indian Pines	81.07±0.15	98.04±0.05	98.97±0.04	94.06±0.07	99.21±0.01	**99.84±0.06**
University of Pavia	87.71± 0.05	98.12±0.33	98.62±0.03	95.27±0.08	99.32±0.02	**99.97±0.01**
Salinas	93.29±0.02	98.79±0.07	98.88±0.06	97.48±0.76	99.06±0.15	**99.48±0.01**
Houston 2018	69.21±1.24	76.59±2.17	78.94±6.06	74.63±3.23	81.08±1.58	**81.96±1.14**

NB: Boldface indicates the highest value obtained.

IV. Ablation Analysis and Discussion

This section elaborates on the various sets of experiments conducted to analyze and validate the impact and effectiveness of each module, as well as the combination of different modules in the proposed framework.

A. Impact of Various Modules on PiCovS's Performance

Each component of the proposed PiCovS plays a critical role that contributes to obtaining the overall performance. The three main modules are the SRN, the SGN, and the SSAN. Table X provides the classification accuracy for the various modules that were either evaluated as a single modular function or as a combined modular unit. It should be noted that the superpixel generation task is set as a submodule to the SGN. From Table X, it can be observed that the combination of the three modules leads to achieving higher performance accuracy than when they were evaluated individually indicating the complementarity among the various modules. However, the SGN module obtained the least OA value for the Houston University-2018 data as against the accuracy it obtained when it was evaluated on the Indian Pines, University of Pavia, and Salinas datasets although combining it with the SSAN module greatly improved the performance. The spectral reduction module (SRN) also aided the proposed PiCovS in obtaining a higher accuracy on all the datasets. The inclusion of the SRN module brings an increase of 0.59%, 0.65%, 0.42%, and 0.88% in classification accuracy on the Indian Pines, University of Pavia, Salinas, and Houston University datasets, respectively, due to the SRN ability to produce refined and optimized initial features for both the pixel-level SSAN and superpixel-based GCN modules. This finding indicates that the SRN can enhance the discrimination of spectral and optimize the features so that the other modules could aid in increasing the accuracy. This is more evident from Table X that, when SGN was singularly evaluated on the Indian Pines data, where fewer spectral interdependencies were captured, the obtained results were not encouraging unless a combination of the SGN and other modules was conducted to improve the performance.

The highest performance rate of 99.84 ± 0.06, 99.97 ± 0.37, 99.48 ± 0.13, and 81.96 ± 1.14 was obtained when all modules were integrated together as a singular modular unit and evaluated on Indian Pines, University of Pavia, Salinas, and Houston University datasets, respectively.

B. Impact of Segmentation Techniques and Segmentation Scale

As shown in Table XI and Fig. 12, the manifold SLIC [57] superpixel segmentation λ is a hyperparameter that we used to examine how different superpixel numbers influence the accuracy of PiCovS. Other superpixel generation algorithms, such as the quick-shift (QS) [58], the Felzenszwalb and Huttenlocher's (FH) technique [59], the linear spectral clustering (LSC) [60], and the SEEDS [61], were employed, and the results were employed in Table XI. The straightforward implementation of PiCovS, makes it easy for PiCovS's experimental configurations to be adaptable, thereby illustrating the its practical application in real-world scenarios.

Applying the modified M-SLIC directly to HSI to calculate the association matrix and adjacency mask might cause degeneration in the segmentation accuracy. For this reason, we perform LDA to reduce the dimensionality of the feature space while preserving the discriminative information to transform the original high-dimensional features into a lower dimensional space where class separability is maximized. The influence of the LDA is examined on all four datasets, and the results are reported in Table XII. The results provided in Table XII demonstrate LDA impacts the performance of PiCovS greatly.

Subsequently, the reduced-dimensional features obtained from LDA are fed as input for the superpixel generation technique (M-SLIC with texture information). We set and analyze λ to 100, 200, 300, 400, and 500, considering that

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TABLE XI

PICOVS OA (%) INDICES WITH DIFFERENT SUPERPIXEL SEGMENTATION TECHNIQUES

Method/Dataset	Conventional SLIC	MSLIC+Texture	FH	LSC	SEEDS
Indian Pines	99.14±0.06	**99.88±0.04**	98.95±0.05	99.78±0.08	99.84±0.07
University of Pavia	99.28±0.03	99.92±0.06	99.43±0.02	99.97±0.01	**99.99±0.0**
Salinas	99.11±0.02	**99.48±0.06**	99.29±0.02	99.35±0.02	99.29±0.03
Houston University-2018	80.99±2.91	**81.96±1.14**	81.42±2.72	81.38±1.02	81.71±1.43

NB: Boldface indicates the highest value obtained.

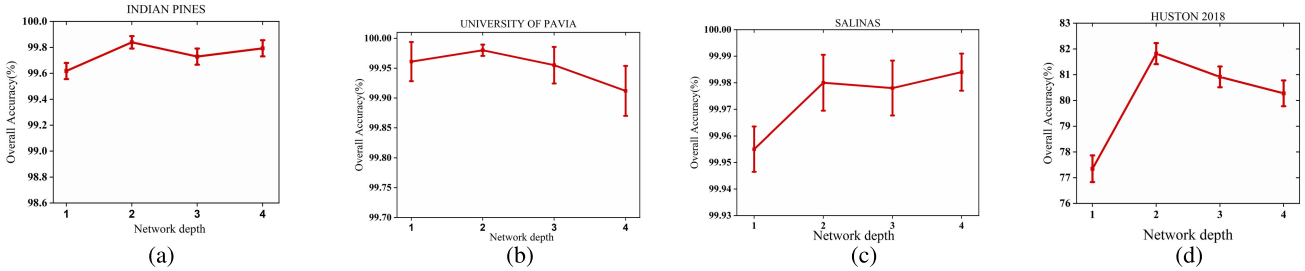


Fig. 11. Performance of PiCovS's classification method with various network depths for each dataset. (a) Indian Pines. (b) Pavia University. (c) Salinas. (d) Houston Univesity-2018.

TABLE XII

OA (%) INDICES OF PICOVS WITH AND WITHOUT THE LDA

Dataset	Without LDA	With LDA
Indian Pines	98.95±0.17	**99.84±0.06**
University of Pavia	89.12±0.13	**99.97±0.01**
Salinas	99.01±1.62	**99.48±0.01**
Houston 2018	80.41±0.12	**81.96±1.14**

NB: Boldface indicates the highest value obtained.

superpixel segmentation is a vital submodule of the superpixel GCN. From Table XI, PiCovS achieved the highest OA score when the MSLIC with texture was used to generate the superpixels on three datasets except for the University of Pavia dataset, where the SEEDS superpixel generation algorithm obtained the highest OA value. Table XI also provides performance values for various superpixel generation techniques, and it is evident that the PiCovS still obtained high accuracy values with various segmentation techniques, demonstrating its robustness and effectiveness. PiCovS yields the highest accuracy when $\lambda = 100$ for all four datasets. It should be noted that smaller λ leads to the generation of more superpixel numbers with less spatial contextual information contained in superpixels, whereas a larger λ indicates fewer superpixel numbers, leading to the inclusion of pixels of different classes in individual superpixel, which degenerates the model's performance.

C. Impact of Network Depth

The network depth is a parameter that impacts the classification accuracy, robustness, and stability of the proposed work. To this end, we set the network depth layers as $L_{SR} = L_{SS} = L_{SG} = [1, 2, 3, 4]$ and then evaluate each depth of the PiCovS to ascertain the optimal network depth. Like all experiments, we repeat each experiment ten times, and their mean values and corresponding standard deviations are provided in Fig. 11.

From Fig. 11, the optimal depth where PiCovS performs best is when network depth is 2 for all four datasets. However, for the University of Pavia dataset, PiCovS obtained a comparable OA when the method was evaluated with a network depth of 1 [see Fig. 11(b)]. One major observation that we made was that the performance of the model degenerates when the network depth is more than two layers, which is mainly due to the oversmoothing ability of the superpixel-based GCN branch. In spite of that, we observe an improvement in performance as the network depth grows for the Salinas dataset. One observation that was made is that the smoothness quality of the superpixel correlates with the network depth, i.e., the smoother the classification map, the more the network layers are needed to obtain higher performance for the Salinas dataset [see Fig. 11(c)]. This is due to the Salinas dataset relatively having large-scale land covers than the remaining three datasets that relatively have small land covers. To enhance robustness and address the issue of over-smoothing, especially for smaller land cover areas, we leverage local detailed information extracted by the pixel-based CNN. This is complemented by a second-order statistical global-attention branch, which captures long-range information and introduces pixel diversity through the learning process.

D. Model Complexity and Execution Time Analysis

Table XIII and Fig. 13 provide the computational complexity in terms of the training and testing times for the various methods computed on the four publicly available datasets. Furthermore, the number of trainable parameters and the floating-point operations (FLOPs) that are used to assess the model complexity of the PiCovS method and the comparable methods are also provided in Table XIV. Similar to previous experiments, we followed the experimental settings by repeating each experiment ten times, and their mean values and their corresponding standard deviation values are reported in Table XIII. As provided in Table XIII and depicted in Fig. 13, the pixel–superpixel-based methods use lesser training time than the CNN-based deep learning methods mainly due to

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TABLE XIII
VARIOUS METHODS' EXECUTION TIMES (IN SECONDS) FOR EACH DATASET

Dataset	Time	DCNN	SSRN	FDSSC	HybridSN	DBDA	WFCG	EMS-GCN	FDGC	CEGCN	PiCovS(Ours)
Indian	Train.	40.2	185.7	111.6	239.8	278.8	53.5	23.02	20.53	**16.3**	22.7
Pines	Test.	5.7	11.6	169	9.7	19.9	1.14	0.6	**0.55**	0.6	0.58
University	Train.	**22.1**	120.8	70.2	97.1s	136.7	172.8	71.05	89.3	163.7	89.6
of Pavia	Test.	30.9	61.2	93.1	110.7	52.6s	6.12	**3.14**	6.4	5.2	4.8
Salinas	Train.	137.6	176.6	142.4	375.1	230.1	85.3	131.06	**27.13**	79.6	70.3
	Test.	14.3	51.0	89.4	46.4	103.8	4.9	7.6	3.5	3.6	**2.3**
Houston 2018	Train.	1055.30	4120.12	3052.57	2022.92	2561.43	684.27	693.11	571.60	703.94	**569.73**
	Test.	59.85	108.31	73.86	82.54	65.39	37.18	35.01	**34.02**	36.75	34.51

TABLE XIV
COMPARISONS IN TERMS OF THE TRAINABLE PARAMETERS AND FLOPS OF VARIOUS DEEP LEARNING-BASED METHODS ON THE UNIVERSITY OF PAVIA DATASET

Complexity	DCNN	SSRN	FDSSC	HybridSN	DBDA	WFCG	EMS-GCN	FDGC	CEGCN	PiCovS
Params	101.81K	355.633K	284.659K	3.94M	592.682K	236.572K	201.695K	184.532K	152.867K	138.539K
FLOPs $\times 10^6$	389.268	449.262	613.849	301.958	692.128	15127.14	15510.55	11684.55M	13094.82	14205.94

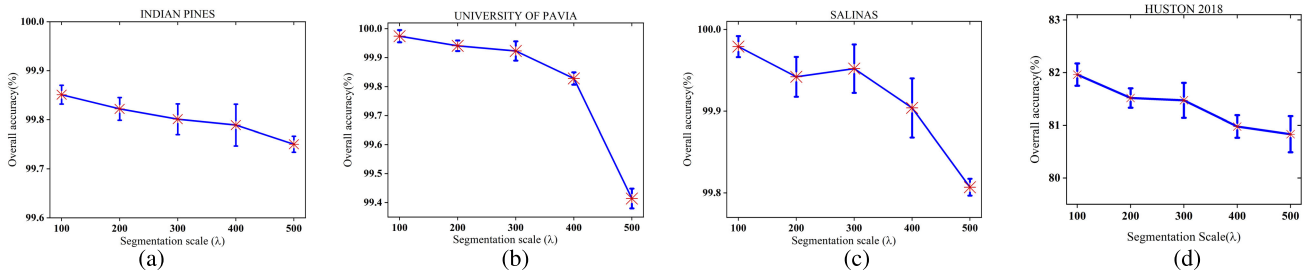


Fig. 12. Impact of various superpixel segmentation scales on the classification performance of PiCovS for each dataset: (a) Indian Pines. (b) Pavia University. (c) Salinas. (d) Houston Univesity-2018.

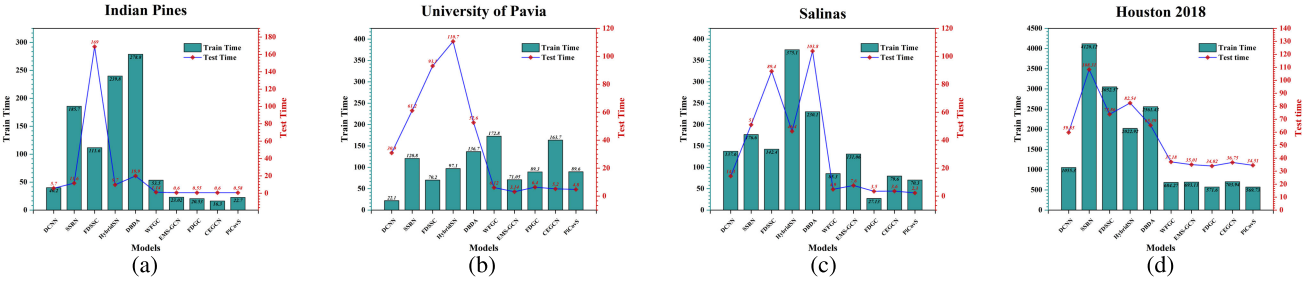


Fig. 13. Comparisons in terms of the training and testing times of different methods. (a) Indian Pines. (b) Pavia University. (c) Salinas. (d) Houston Univesity-2018.

the extra computation. CEGCN used 16.3 s to train the model on the Indian Pines dataset; however, the FDGC method only lasted for 0.55 s for testing when evaluated on the Indian Pines. Exceptionally, the DCNN only lasted 22.1 s during training on the University of Pavia datasets with the EMS-GCN lasting for 3.14 s during inference. PiCovS in terms of training obtained the fastest time when evaluated on the Houston University dataset. During inference, PiCovS provided the fastest testing time of 2.3 s on the Salinas dataset. Other than these, the training times and testing time for the PiCovS on the Indian Pines and the University of Pavia datasets are comparable to other state-of-the-art methods. It should be noted that the superpixel generator's executing time is added to the operating time in our technique. While the execution times may not be significantly faster than other methods on the Indian Pines and the University of Pavia datasets, it is essential to emphasize that our primary objective was to enhance the accuracy and performance of the classification, which we achieved through the fusion of pixel-level and superpixel-level features and the introduction of second-order statistical representations.

On computational complexity, as reported in Table XIV, the FDGC, CEGCN, and DCNN aside from PiCovS require fewer trainable parameters compared with other methods, while their FLOPs, which are computed at the first batch, are 1000 times higher than all the CNN-based methods. The FLOPs are higher mainly due to the fact that the pixel–superpixel-based hybrid methods are directly fed the whole image as input instead of patches into the models for each batch. The proposed method PiCovS has a smaller number of parameters compared to most comparable methods except for the DCNN method, which has fewer parameters than the proposed PiCovS. In terms of FLOPs, the proposed PiCovS has nearly 14205 $\times$ 10^6.

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1) Performance Under Limited Samples: In industrial applications, HSIs are usually expensive and demand a lot of time to accurately label pixels as expertise supervision is required, limiting the power and application of deep learning models. Therefore, it is imperative to build data-efficient models that exhibit robustness and effectiveness, whether dealing with a limited or extensive number of data samples. Fig. 10 provides the effect of training with a different number of training samples on the proposed PiCovS and other state-of-the-art methods. We investigated by evaluating the performance of different methods with varying amounts of training samples that were randomly chosen from five to 25 samples per class as training samples in each experiment on each dataset. The mean OA indices of several methods under various sizes of training samples are shown in Fig. 10 after ten repetitions of each experiment. With deep learning models performing better when trained with large amounts of training samples, it is not surprising that the OA of all the methods increases when a large number of training samples per class is used. However, for the University of Pavia and Salinas datasets, it can be observed the proposed method obtained higher OA values than when it was evaluated on the Indian Pines and Houston University datasets although the proposed PiCovS outperformed all comparable methods. Using only five training samples per class, the proposed PiCovS gained more than 79%, 92%, 98%, and 65% on Indian Pines, University of Pavia, Salinas, and Houston University datasets, respectively. This demonstrates that the proposed PiCovS method is robust and has excellent small-sample learning capabilities.

V. Conclusion

In this study, we have presented PiCovS, a novel deep-learning network for HSI classification that effectively fuses the power of CNNs and GCNs. By combining CNN-based local pixel-level features with GCN-based superpixel-level features, PiCovS achieves a comprehensive understanding of both small-regular and large non-Euclidean image regions. To address the structural incompatibility between CNN and GCN data representations, we introduced a graph encoder–decoder, facilitating the propagation of features between superpixel-based graph nodes and image pixels. This adaptive approach enables the GCN to effectively capture relevant features based on the specific HSI content, enhancing classification accuracy.

Furthermore, we introduced novel second-order statistical mechanisms throughout the network, promoting diversity among neighboring pixels and capturing long-range spectral–spatial dependencies. By emphasizing significant features and suppressing less relevant ones, our approach enhances the discriminative power of the model, resulting in improved classification performance.

Extensive experiments on four HSI datasets validated the superiority of PiCovS over other state-of-the-art deep-learning-based HSI classification methods. PiCovS demonstrated the best inference accuracy and comparable time in most experiments and exhibited superior generalization with limited training samples, achieving higher overall accuracies and better convergence compared to other counterparts.

While our experimental results set new state-of-the-art standards, we acknowledge that the GCN module tested separately did not achieve competitive scores compared to the other modules. Therefore, for future work, we aim to explore more efficient graph learning techniques tailored specifically for HSI classification, potentially employing multiscale-purpose algorithms. In addition, we plan to further investigate and extend the second-order statistical mechanism used in the GCN to better model adjacent land covers and improve the overall performance of the proposed approach.

In conclusion, PiCovS offers a robust and innovative solution for HSI classification by effectively integrating CNN and GCN features and leveraging second-order statistical mechanisms. We believe that our method has great potential for real-world applications in remote sensing and related fields, contributing to the advancement of HSI analysis.

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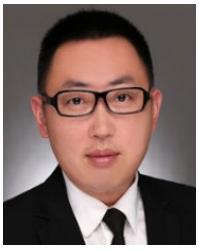
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